

**EC-Council**



**Digital Forensics Essentials**



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# MODULE 5

## Windows, Linux and Mac Forensics



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# Windows Forensics

## SUBTOPIC 1



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## Module Objectives

- 1 Understanding the Collection Of Volatile and Non-volatile Information
- 2 Understanding the Windows Memory and Registry Analysis
- 3 Understanding How to Examine Cache, Cookie, and History Recorded in Web Browsers
- 4 Understanding How to Examine Windows Files and Metadata



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## Module Flow

1

**Collect Volatile and Non-Volatile Information**

2

**Perform Windows Memory and Registry Analysis**

3

**Examine Cache, Cookie, and History Recorded in Web Browsers**

4

**Examine Windows Files and Metadata**



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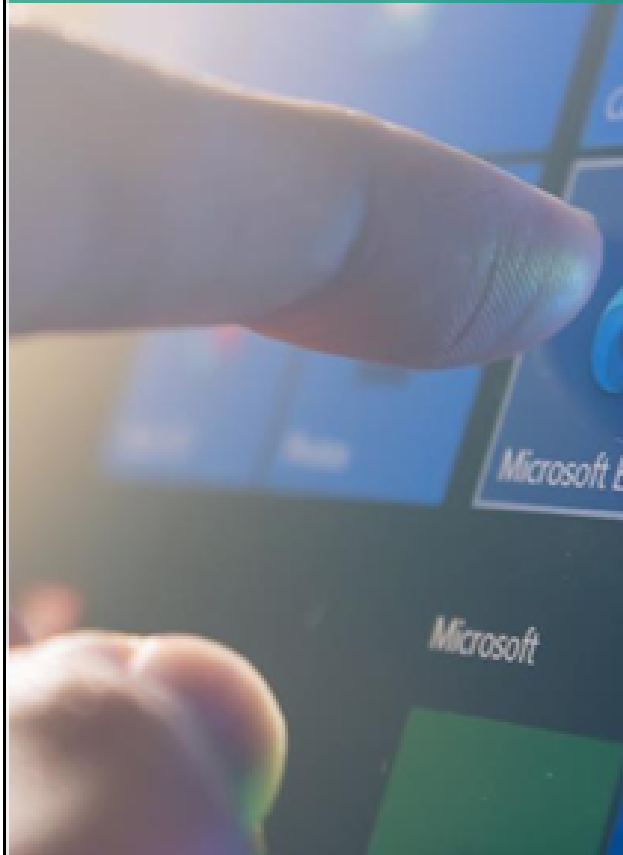
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# Introduction to OS Forensics



**Windows, Mac, and Linux** are the three most widely used operating systems (OSes). Thus, the probability for an investigator to come across these OSes at the crime scene is very high.



Performing OS forensics to **uncover the underlying evidence** is a challenging task as it requires the investigator to have thorough knowledge of these OSes



To conduct a successful **digital forensic examination** in Windows, Mac, and Linux, one should be **familiar with** their **working, commands** or **methodologies**, in order to be able to extract volatile and non-volatile data with OS-specific tools



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# Collecting Volatile Information



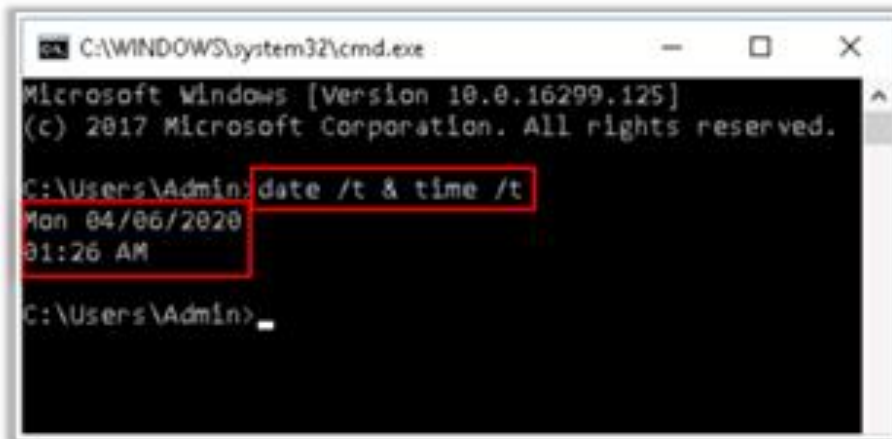
- ❑ **Volatile information** can be easily modified or lost when the system is shut down or rebooted
- ❑ Collecting volatile information helps determine **a logical timeline of the security incident** and the responsible users
- ❑ **Volatile data** resides in registers, cache, and RAM

## Volatile information includes:

- System time
- Logged-on user(s)
- Network information
- Open files
- Network connections
- Network status
- Process information
- Process-to-port mapping
- Process memory
- Mapped drives
- Shares
- Clipboard contents
- Service/driver information
- Command history

# Collecting System Time

- ❑ It provides details of the **information collected** during the **investigation**
- ❑ It helps in re-creating the **accurate timeline** of events that occurred on the system
- ❑ System uptime provides an idea of when an **exploit attempt** might have been successful



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.16299.125]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\Users\Admin>date /t & time /t
Mon 04/06/2020
01:26 AM
C:\Users\Admin>
```

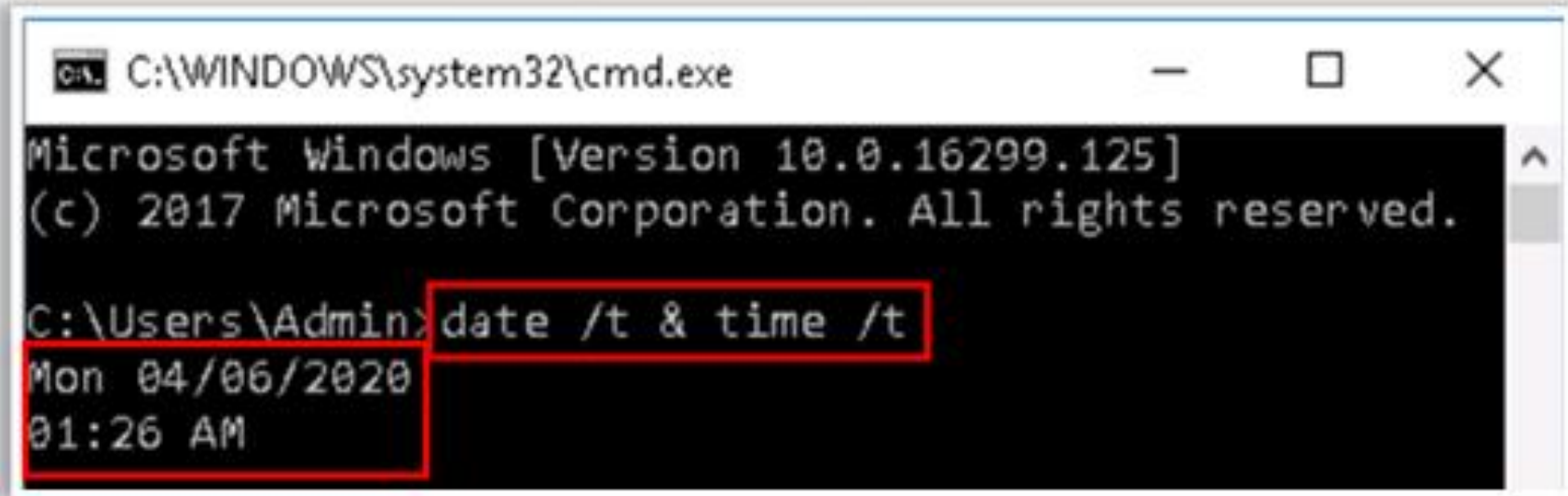
**Note:** Acquire or duplicate the memory of the target system before extracting volatile data, as the commands used in the process can alter contents of media and make the proof legally invalid





## Command:

```
date /t & time /t
```



A screenshot of a Windows Command Prompt window. The title bar shows the path 'C:\WINDOWS\system32\cmd.exe'. The window contains the following text: 'Microsoft Windows [Version 10.0.16299.125]', '(c) 2017 Microsoft Corporation. All rights reserved.', and the command prompt 'C:\Users\Admin>'. The command 'date /t & time /t' has been entered and is highlighted with a red rectangular box. Below the command, the output is displayed: 'Mon 04/06/2020' and '01:26 AM', both of which are also highlighted with red rectangular boxes.

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.16299.125]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\Users\Admin>date /t & time /t
Mon 04/06/2020
01:26 AM
```



# Collecting Logged-On Users

## PsLoggedOn

PsLoggedOn is an applet that **displays** both the **users logged on** locally and via resources for either on the local, or a remote computer



### Syntax:

```
psloggedon [- ] [-l] [-x] [\\computername | username]
```

```
Administrator: C:\Windows\system32\cmd.exe

C:\Users\Tania\OneDrive\Forensics\Acquiring Volatile Information\PSTools>
PsLoggedon64.exe

PsLoggedon v1.35 - See who's logged on
Copyright (C) 2000-2016 Mark Russinovich
Sysinternals - www.sysinternals.com

Users logged on locally:
    2/25/2020 10:49:16 PM    WIN-HUQ0C24IC1\Administrator

Users logged on via resource shares:
    2/27/2020 8:42:56 PM    (null)\Administrator
```



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## Collecting Logged-On Users (Cont'd)

- ❑ net session command displays **computer** and **usernames** on a server, open files, and duration of sessions

**net sessions Command**



**Syntax:**

```
net sessions  
[\\<ComputerName>]  
[/delete] [/list]
```



```
Administrator: Command Prompt
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Windows\system32>net sessions

Computer          User name          Client Type        Opens Idle time
-----
\\192.168.0.137    Administrator      4 00:00:26
\\192.168.0.178    Administrator      2 00:07:24
The command completed successfully.

C:\Windows\system32>
```

## Collecting Logged-On Users (Cont'd)

- ❑ net session command displays **computer** and **usernames** on a server, open files, and duration of sessions

**net sessions Command**



**Syntax:**

```
net sessions  
[\\<ComputerName>]  
[/delete] [/list]
```



```
Administrator: Command Prompt
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Windows\system32>net sessions

Computer                User name                Client Type              Opens Idle time
-----
\\192.168.0.137          Administrator             4 00:00:26
\\192.168.0.178          Administrator             2 00:07:24
The command completed successfully.

C:\Windows\system32>
```

# Collecting Open Files: net file Command



Collect **information about the files opened** by the intruder using remote login

## net file command

- ❑ Displays **details of open shared files on a server**, such as a name, ID, and the number of each file locks, if any. It also closes individually shared files and removes file locks.
- ❑ The syntax of the net file command:

**net file [ID [/close]]**



```
Administrator: Command Prompt
C:\Windows\system32>net file

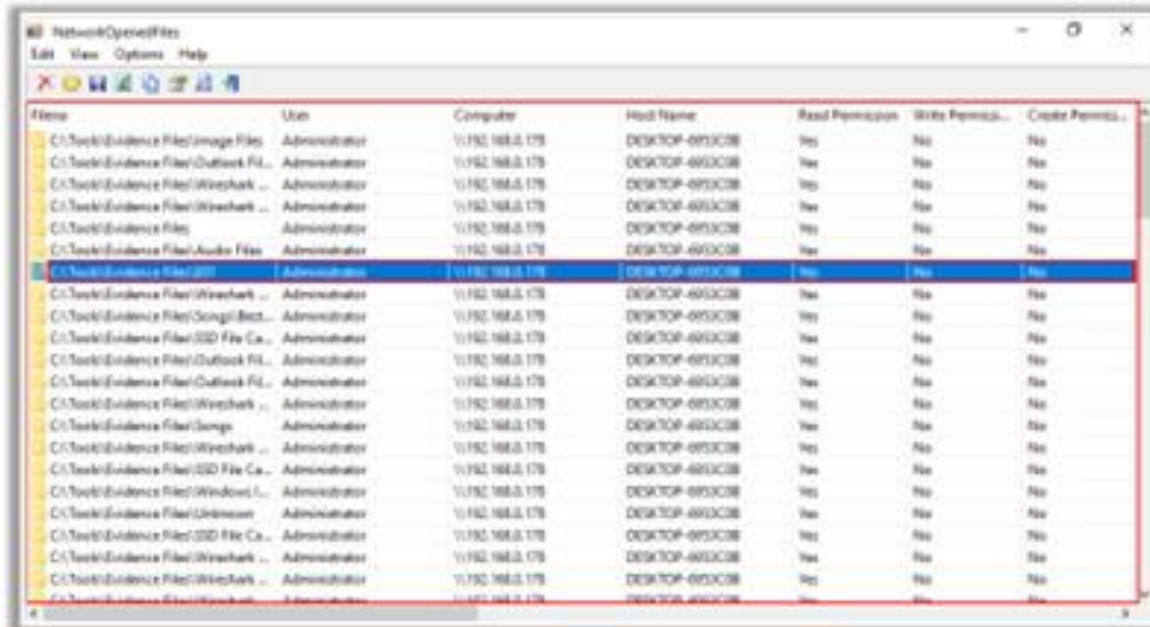
ID            Path                                User name        # Locks
-----
112           C:\Tools\Evidence Files            Administrator    0
113           C:\Tools\Evidence Files\ADS Files  Administrator    0
114           C:\Tools\                          Administrator    0
132           C:\Tools\                          Administrator    0
The command completed successfully.
```





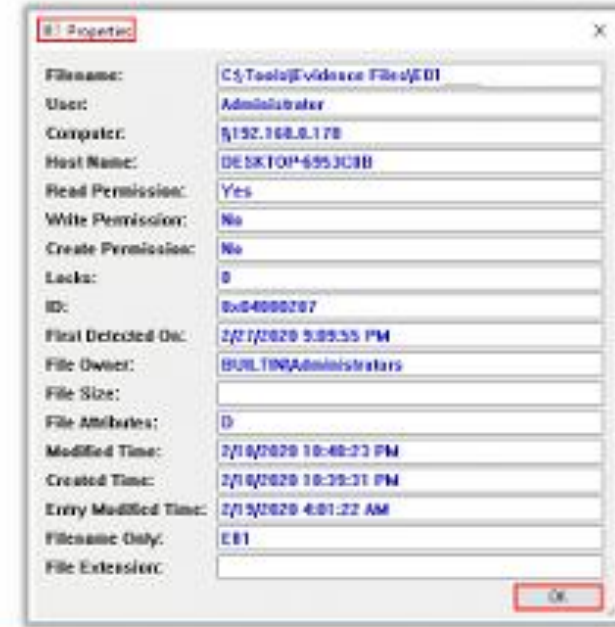
# Collecting Open Files: Using NetworkOpenedFiles

- ❑ **NetworkOpenedFiles** is a utility for Windows OS that lists all the files currently **opened on the host system** through remote login
- ❑ It displays the Filename, Computer and Username, Permission information (Read/Write/Create), Locks count, File Size, File Attributes, etc.



The screenshot shows the NetworkOpenedFiles application window. It contains a table with the following columns: File, User, Computer, Host Name, Read Permission, Write Permission, and Create Permission. The table lists various files and folders, with the file 'C:\Tools\Evidence Files\ED1' highlighted in blue.

| File                                   | User          | Computer      | Host Name       | Read Permission | Write Permission | Create Permission |
|----------------------------------------|---------------|---------------|-----------------|-----------------|------------------|-------------------|
| C:\Tools\Evidence Files\Image Files    | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Outlook Fil... | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files                | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Audio Files    | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\ED1            | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Songs\Best...  | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\SSD File Ca... | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Outlook Fil... | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Outlook Fil... | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Songs          | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\SSD File Ca... | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Windows L...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Unknown        | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\SSD File Ca... | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |
| C:\Tools\Evidence Files\Winchark ...   | Administrator | 192.168.0.178 | DESKTOP-6953C8B | Yes             | No               | No                |



The screenshot shows the Properties dialog box for the file 'C:\Tools\Evidence Files\ED1'. The fields are filled with the following information:

- Filename: C:\Tools\Evidence Files\ED1
- User: Administrator
- Computer: 192.168.0.178
- Host Name: DESKTOP-6953C8B
- Read Permission: Yes
- Write Permission: No
- Create Permission: No
- Locks: 0
- ID: 6x0400207
- First Detected On: 2/17/2020 3:05:55 PM
- File Owner: BUILTIN\Administrators
- File Size:
- File Attributes: D
- Modified Time: 2/16/2020 10:40:23 PM
- Created Time: 2/16/2020 10:39:31 PM
- Every Modified Time: 2/15/2020 4:01:22 AM
- Filename Only: ED1
- File Extension:

| NetworkOpenedFiles                     |               |                 |                 |                 |                   |                   |
|----------------------------------------|---------------|-----------------|-----------------|-----------------|-------------------|-------------------|
| Edit View Options Help                 |               |                 |                 |                 |                   |                   |
|                                        |               |                 |                 |                 |                   |                   |
| File Name                              | User          | Computer        | Host Name       | Read Permission | Write Permissi... | Create Permiss... |
| C:\Tools\Evidence Files\Image Files    | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Outlook Fil... | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files                | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Audio Files    | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\E01            | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Songs\Best...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\SSD File Ca... | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Outlook Fil... | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Outlook Fil... | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Songs          | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\SSD File Ca... | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Windows L...   | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Unknown        | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\SSD File Ca... | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |
| C:\Tools\Evidence Files\Wireshark ...  | Administrator | \\192.168.0.178 | DESKTOP-6953C08 | Yes             | No                | No                |

 Properties ✕

|                      |                                  |
|----------------------|----------------------------------|
| Filename:            | C:\Tools\Evidence Files\E01_____ |
| User:                | Administrator                    |
| Computer:            | \\192.168.0.178                  |
| Host Name:           | DESKTOP-6953C0B                  |
| Read Permission:     | Yes                              |
| Write Permission:    | No                               |
| Create Permission:   | No                               |
| Locks:               | 0                                |
| ID:                  | 0x04000207                       |
| First Detected On:   | 2/27/2020 9:09:55 PM             |
| File Owner:          | BUILTIN\Administrators           |
| File Size:           |                                  |
| File Attributes:     | D                                |
| Modified Time:       | 2/18/2020 10:40:23 PM            |
| Created Time:        | 2/18/2020 10:39:31 PM            |
| Entry Modified Time: | 2/19/2020 4:01:22 AM             |
| Filename Only:       | E01                              |
| File Extension:      |                                  |

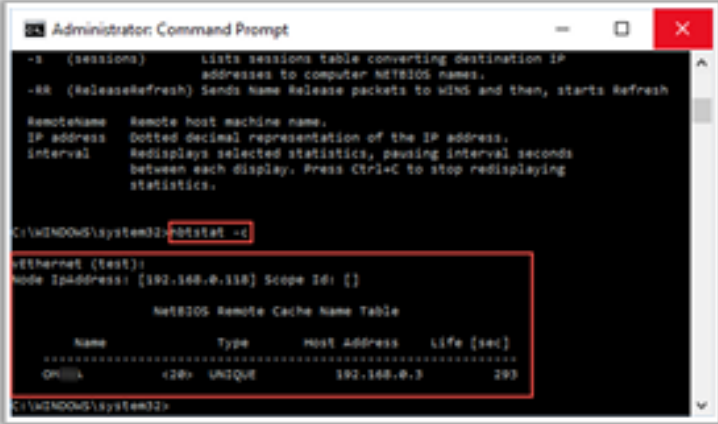
OK

## Collecting Network Information

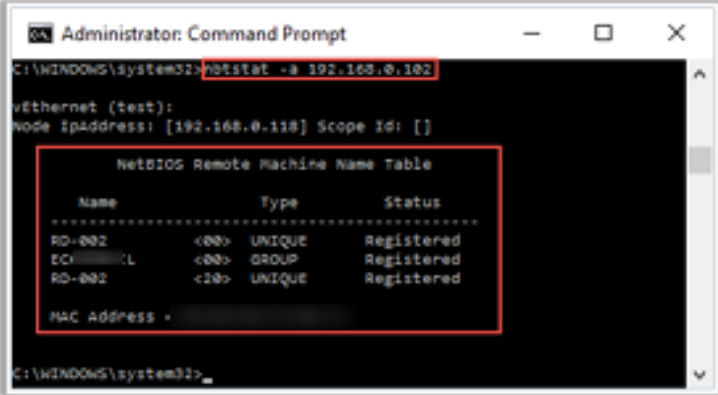
- ❑ Intruders after gaining access to a remote system, try to **discover other systems** that are available on the network
- ❑ NetBIOS name table cache **maintains a list of connections** made to other systems using NetBIOS
- ❑ The Windows inbuilt command line utility **nbtstat** can be used to view NetBIOS name table cache
- ❑ The **nbtstat -c** option shows the contents of the NetBIOS name cache, which contains NetBIOS name-to-IP address mappings

### Syntax:

```
nbtstat [-a RemoteName] [-A IP address]  
[-c] [-n] [-r] [-R] [-RR] [-s] [-S]  
[interval]
```



```
Administrator: Command Prompt  
-s (sessions) Lists sessions table converting destination IP  
addresses to computer NetBIOS names.  
-RR (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh  
  
RemoteName Remote host machine name.  
IP address Dotted decimal representation of the IP address.  
Interval Redisplay selected statistics, pausing interval seconds  
between each display. Press Ctrl+C to stop redisplaying  
statistics.  
  
C:\WINDOWS\system32>nbtstat -c  
  
vEthernet (test):  
Node IpAddress: [192.168.0.118] Scope Id: []  
  
NetBIOS Remote Cache Name Table  
  
Name Type Host Address Life (sec)  
-----  
00000000 <20> UNIQUE 192.168.0.3 293  
  
C:\WINDOWS\system32>
```



```
Administrator: Command Prompt  
C:\WINDOWS\system32>nbtstat -a 192.168.0.102  
  
vEthernet (test):  
Node IpAddress: [192.168.0.118] Scope Id: []  
  
NetBIOS Remote Machine Name Table  
  
Name Type Status  
-----  
RD-002 <00> UNIQUE Registered  
EC-00000000 <00> GROUP Registered  
RD-002 <20> UNIQUE Registered  
  
MAC Address =  
  
C:\WINDOWS\system32>
```



```
Administrator: Command Prompt

-s (sessions) Lists sessions table converting destination IP
               addresses to computer NETBIOS names.
-RR (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

RemoteName Remote host machine name.
IP address Dotted decimal representation of the IP address.
interval Redisplays selected statistics, pausing interval seconds
           between each display. Press Ctrl+C to stop redisplaying
           statistics.

C:\WINDOWS\system32>nbtstat -c

vEthernet (test):
Node IpAddress: [192.168.0.118] Scope Id: []

NetBIOS Remote Cache Name Table

  Name                Type        Host Address    Life [sec]
-----
OM <20> UNIQUE        192.168.0.3     293

C:\WINDOWS\system32>
```



```
Administrator: Command Prompt
C:\WINDOWS\system32>nbtstat -a 192.168.0.102

vEthernet (test):
Node IpAddress: [192.168.0.118] Scope Id: []

NetBIOS Remote Machine Name Table

Name                Type               Status
-----
RD-002              <00>  UNIQUE           Registered
EC( ) :L            <00>  GROUP            Registered
RD-002              <20>  UNIQUE           Registered

MAC Address :

C:\WINDOWS\system32>
```

# Collecting Information about Network Connections

- ❑ Collecting information about the network connections running to and from the victim system allows to locate logged attacker, IRCbot communication, worms logging into Command and Control server
- ❑ **Netstat** with **-ano switch** displays details of the TCP and UDP network connections including listening ports, and the identifiers

## Syntax:

**netstat [-a] [-e] [-n] [-o] [-p <Protocol>] [-r] [-s] [<Interval>]**



```
Administrator Command Prompt
Microsoft Windows [Version 10.0.18546]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>netstat -ano

Active Connections

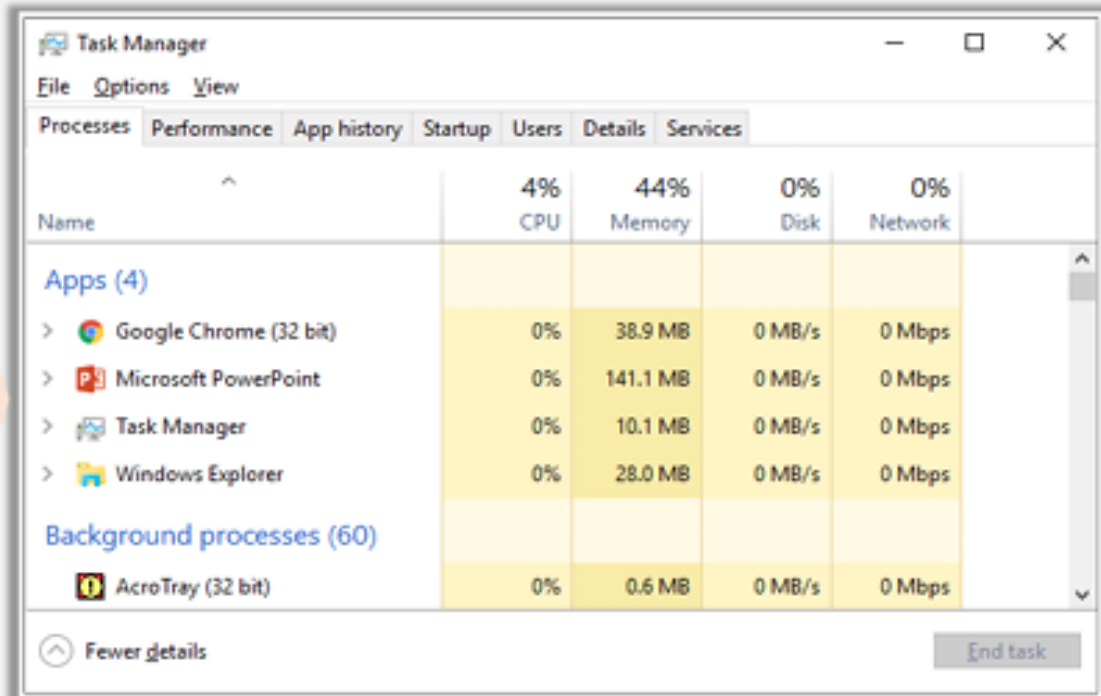
Proto Local Address           Foreign Address         State       PID
TCP   0.0.0.0:135              0.0.0.0:0               LISTENING   928
TCP   0.0.0.0:445              0.0.0.0:0               LISTENING   4
TCP   0.0.0.0:1536             0.0.0.0:0               LISTENING   636
TCP   0.0.0.0:1537             0.0.0.0:0               LISTENING   980
TCP   0.0.0.0:1538             0.0.0.0:0               LISTENING   1152
TCP   0.0.0.0:1539             0.0.0.0:0               LISTENING   1996
TCP   0.0.0.0:1540             0.0.0.0:0               LISTENING   776
TCP   0.0.0.0:1541             0.0.0.0:0               LISTENING   768
TCP   0.0.0.0:2179             0.0.0.0:0               LISTENING   2388
TCP   0.0.0.0:3389             0.0.0.0:0               LISTENING   540
TCP   0.0.0.0:22350            0.0.0.0:0               LISTENING   2340
TCP   0.0.0.0:26143            0.0.0.0:0               LISTENING   4
TCP   127.0.0.1:845            0.0.0.0:0               LISTENING   5272
TCP   127.0.0.1:17600          0.0.0.0:0               LISTENING   5272
TCP   127.0.0.1:26846         127.0.0.1:26847         ESTABLISHED 5272
TCP   127.0.0.1:26847         127.0.0.1:26846         ESTABLISHED 5272
```

# Process Information

- ❑ Investigate the **processes running on a potentially compromised system** and collect the information

**Tools and commands used to collect detailed process information include:**

- ❑ **Task Manager** displays the programs, processes, and services that are currently running on computer



The screenshot shows the Windows Task Manager window with the 'Processes' tab selected. It displays a table of running applications and background processes, including their names, CPU usage, memory usage, disk usage, and network usage.

| Name                             | CPU | Memory   | Disk   | Network |
|----------------------------------|-----|----------|--------|---------|
| <b>Apps (4)</b>                  |     |          |        |         |
| > Google Chrome (32 bit)         | 0%  | 38.9 MB  | 0 MB/s | 0 Mbps  |
| > Microsoft PowerPoint           | 0%  | 141.1 MB | 0 MB/s | 0 Mbps  |
| > Task Manager                   | 0%  | 10.1 MB  | 0 MB/s | 0 Mbps  |
| > Windows Explorer               | 0%  | 28.0 MB  | 0 MB/s | 0 Mbps  |
| <b>Background processes (60)</b> |     |          |        |         |
| AcroTray (32 bit)                | 0%  | 0.6 MB   | 0 MB/s | 0 Mbps  |

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

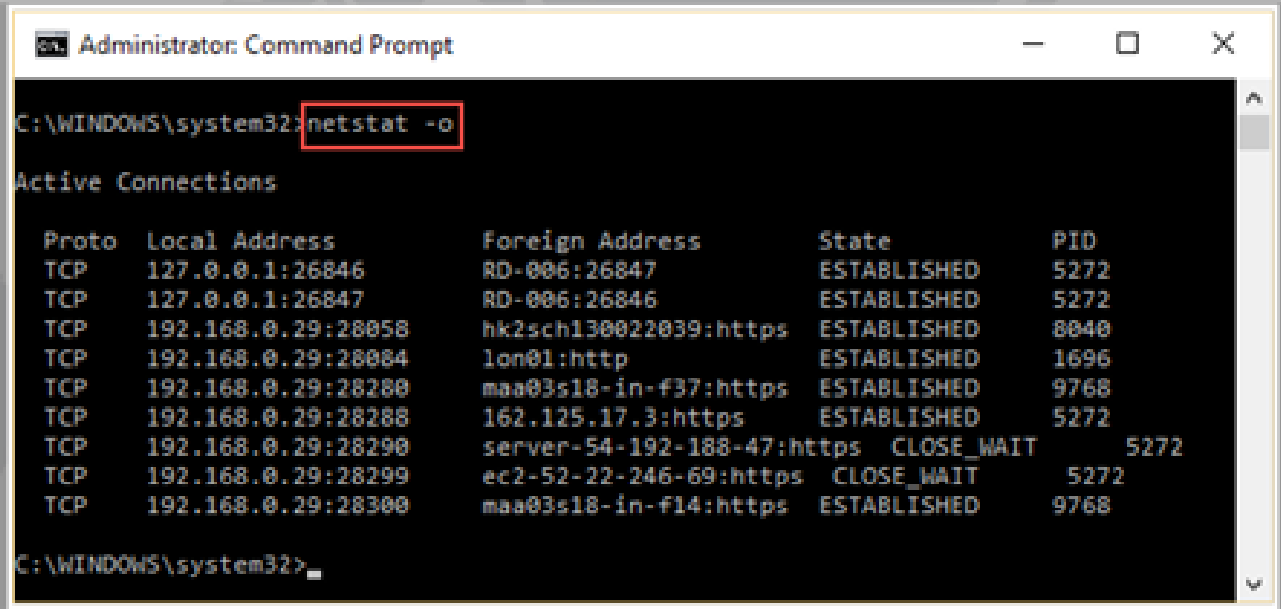
| Name                      | 4%<br>CPU | 44%<br>Memory | 0%<br>Disk | 0%<br>Network |
|---------------------------|-----------|---------------|------------|---------------|
| Apps (4)                  |           |               |            |               |
| > Google Chrome (32 bit)  | 0%        | 38.9 MB       | 0 MB/s     | 0 Mbps        |
| > Microsoft PowerPoint    | 0%        | 141.1 MB      | 0 MB/s     | 0 Mbps        |
| > Task Manager            | 0%        | 10.1 MB       | 0 MB/s     | 0 Mbps        |
| > Windows Explorer        | 0%        | 28.0 MB       | 0 MB/s     | 0 Mbps        |
| Background processes (60) |           |               |            |               |
| ! AcroTray (32 bit)       | 0%        | 0.6 MB        | 0 MB/s     | 0 Mbps        |

^ Fewer details End task

# Process-to-Port Mapping

- ❑ Process-to-port mapping **traces** the **port** used by a **process**, and protocol connected to the IP
- ❑ Tools and commands to retrieve the process-to-port mapping:

>\_ Syntax:  
**netstat -a -n -o**



```
Administrator: Command Prompt
C:\WINDOWS\system32>netstat -o

Active Connections

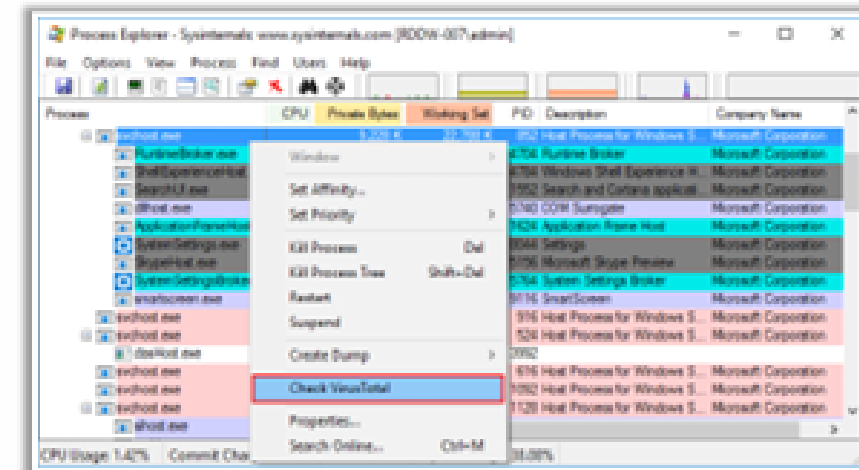
Proto Local Address          Foreign Address         State       PID
TCP   127.0.0.1:26846         RD-886:26847           ESTABLISHED 5272
TCP   127.0.0.1:26847         RD-886:26846           ESTABLISHED 5272
TCP   192.168.0.29:28058      hk2sch130022039:https  ESTABLISHED 8040
TCP   192.168.0.29:28084      lon01:http             ESTABLISHED 1696
TCP   192.168.0.29:28280      maa03s18-in-f37:https  ESTABLISHED 9768
TCP   192.168.0.29:28288      162.125.17.3:https     ESTABLISHED 5272
TCP   192.168.0.29:28290      server-54-192-188-47:https CLOSE_WAIT  5272
TCP   192.168.0.29:28299      ec2-52-22-246-69:https CLOSE_WAIT  5272
TCP   192.168.0.29:28300      maa03s18-in-f14:https  ESTABLISHED 9768

C:\WINDOWS\system32>
```

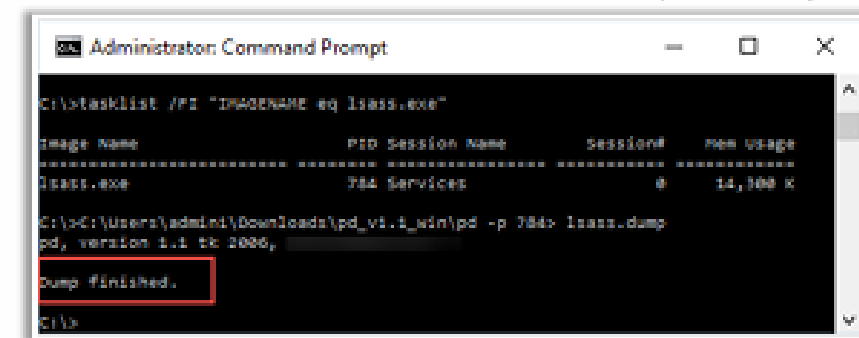


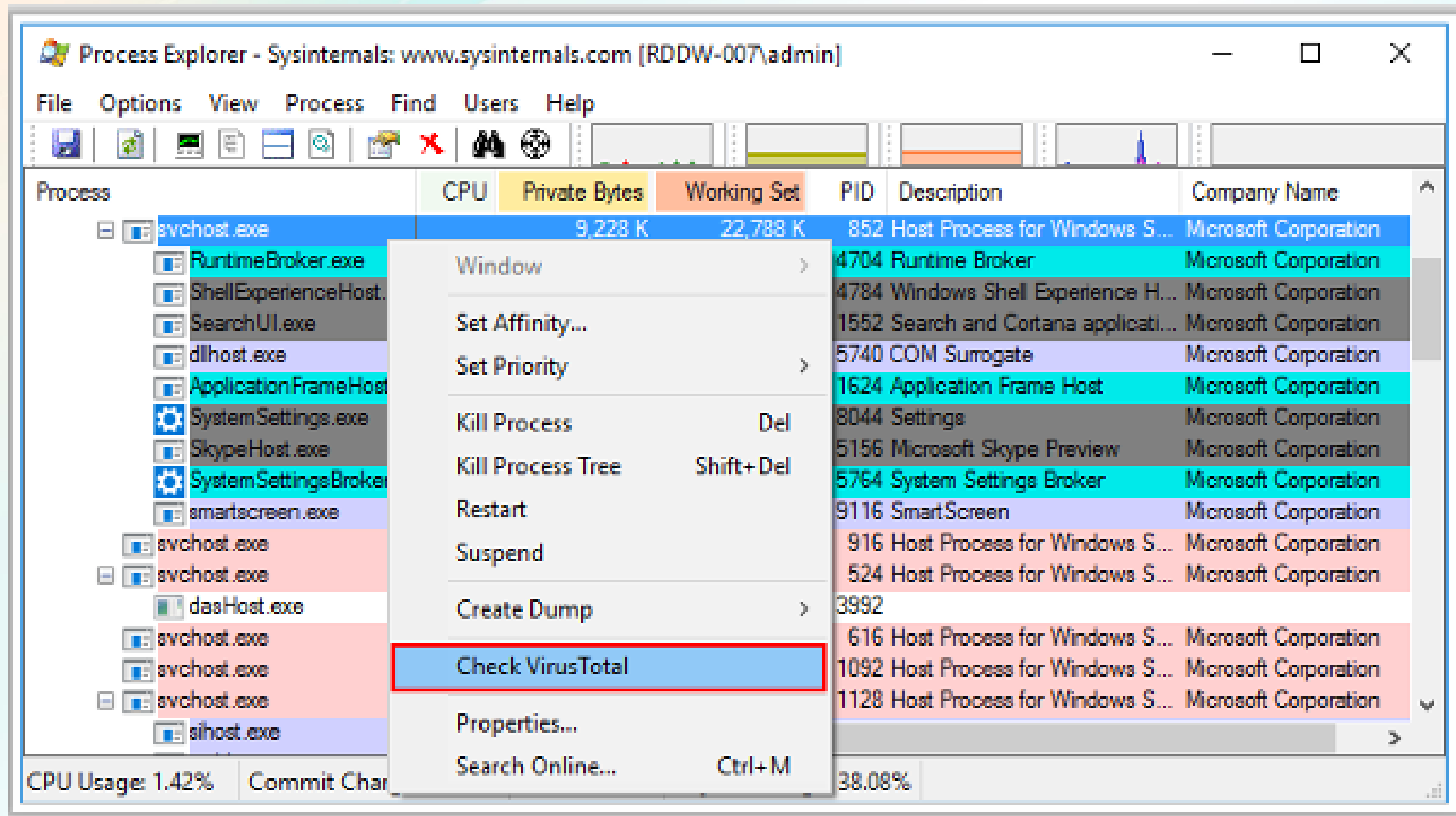
# Examining Process Memory

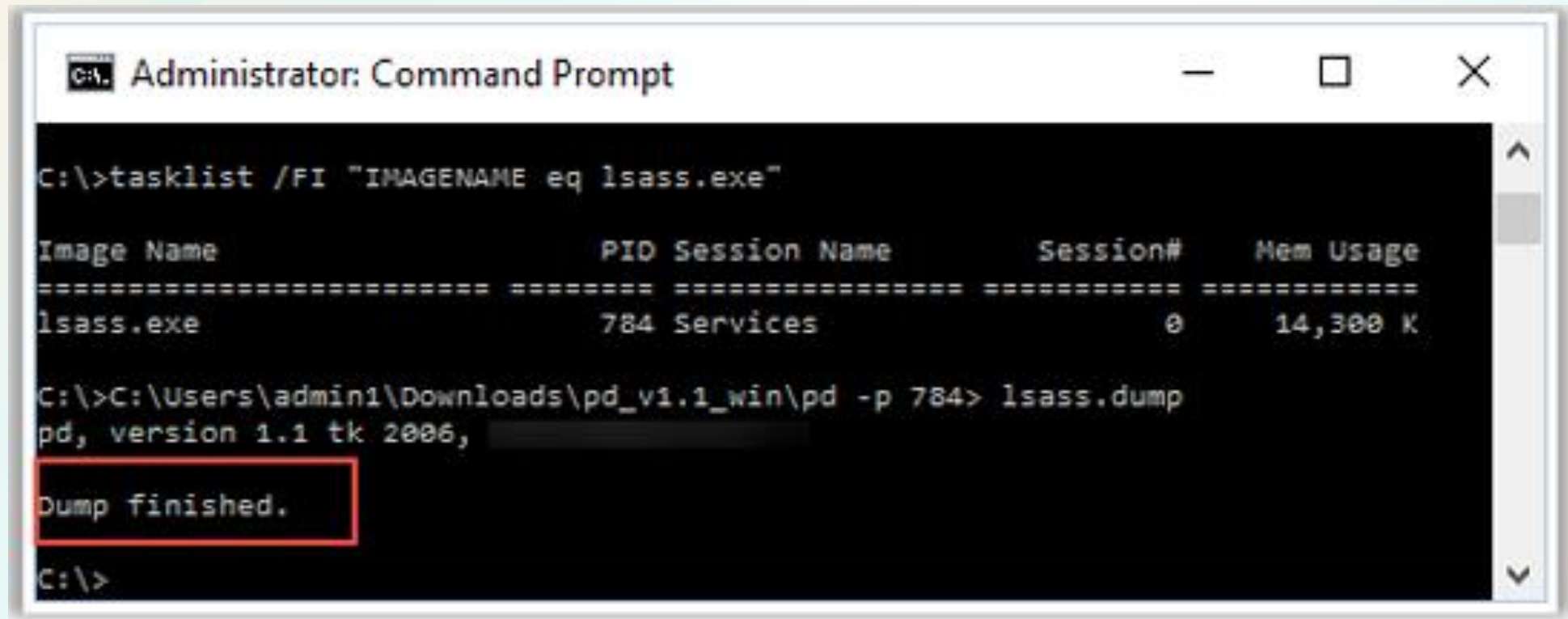
- ❑ Running processes could be **suspicious** or **malicious** in nature
- ❑ **Process Explorer** can be used to check if the process is malicious/suspicious
- ❑ Process Explorer shows information about opened or loaded **handles** and **DLLs** processes
- ❑ If the process is suspicious, it gathers more information by dumping the memory used by the process using tools such as **ProcDump** and **Process Dumper**
- ❑ The tool comes with built-in VirusTotal support to check whether the running process is malicious



<https://docs.microsoft.com>







The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The user has entered the command `tasklist /FI "IMAGENAME eq lsass.exe"`, which displays a table of running processes. The table has five columns: Image Name, PID, Session Name, Session#, and Mem Usage. The only process listed is `lsass.exe` with PID 784, Session Name Services, Session# 0, and Mem Usage 14,300 K. The user then enters the command `C:\Users\admin1\Downloads\pd_v1.1_win\pd -p 784> lsass.dump`. The output shows the version of the tool and the path to the dump file. Finally, the text "Dump finished." is displayed and highlighted with a red rectangle.

```
C:\>tasklist /FI "IMAGENAME eq lsass.exe"

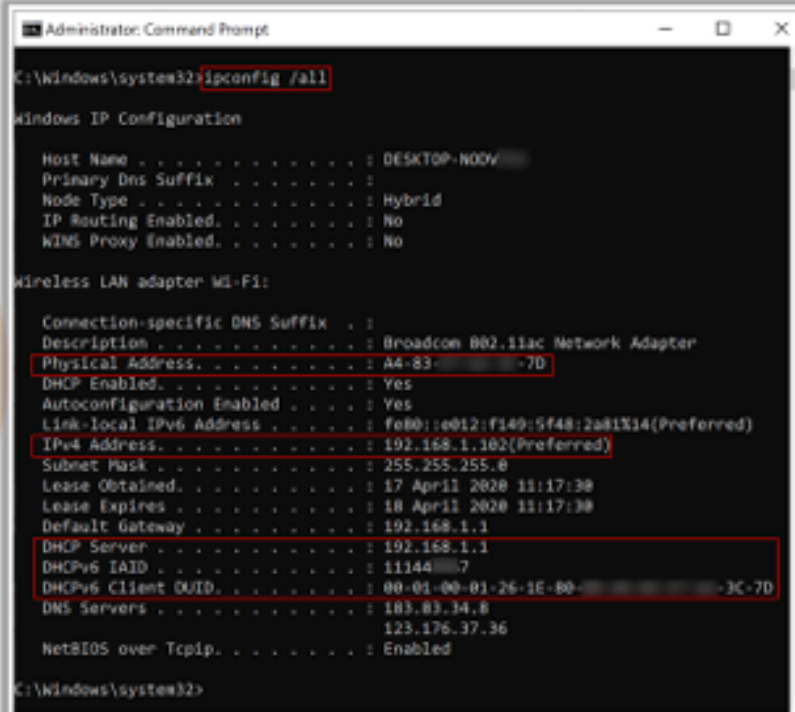
Image Name                               PID Session Name        Session#    Mem Usage
=====
lsass.exe                                784 Services             0         14,300 K

C:\>C:\Users\admin1\Downloads\pd_v1.1_win\pd -p 784> lsass.dump
pd, version 1.1 tk 2006, 
Dump finished.
C:\>
```

Using Process Dumper to dump the memory of a running process

# Collecting Network Status

- ❑ Collect information of the **network interface cards** (NICs) of a system to know whether the system is connected to a **wireless access point** and what **IP address** is being used
- ❑ Tools for the network status detection are:
  - **ipconfig** command
  - **PromiscDetect** tool
  - **Promqry** tool
- ❑ **Ipconfig.exe** is a utility native to Windows systems that displays information about NICs and their status
- ❑ **Ipconfig /all** command displays the network configuration of the NICs on the system



```
Administrator: Command Prompt
C:\Windows\system32>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-W00V
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Broadcom 802.11ac Network Adapter
Physical Address. . . . . : A4-83-7D-
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e012:f149:5f48:2a81%14(Preferred)
IPv4 Address. . . . . : 192.168.1.102(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 17 April 2020 11:17:30
Lease Expires . . . . . : 18 April 2020 11:17:30
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 111447
DHCPv6 Client DUID. . . . . : 00-01-00-01-26-1E-80-3C-7D
DNS Servers . . . . . : 193.83.34.8
123.176.37.36
NetBIOS over Tcpip. . . . . : Enabled

C:\Windows\system32>
```

```
Administrator: Command Prompt

C:\Windows\system32>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-NODV
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

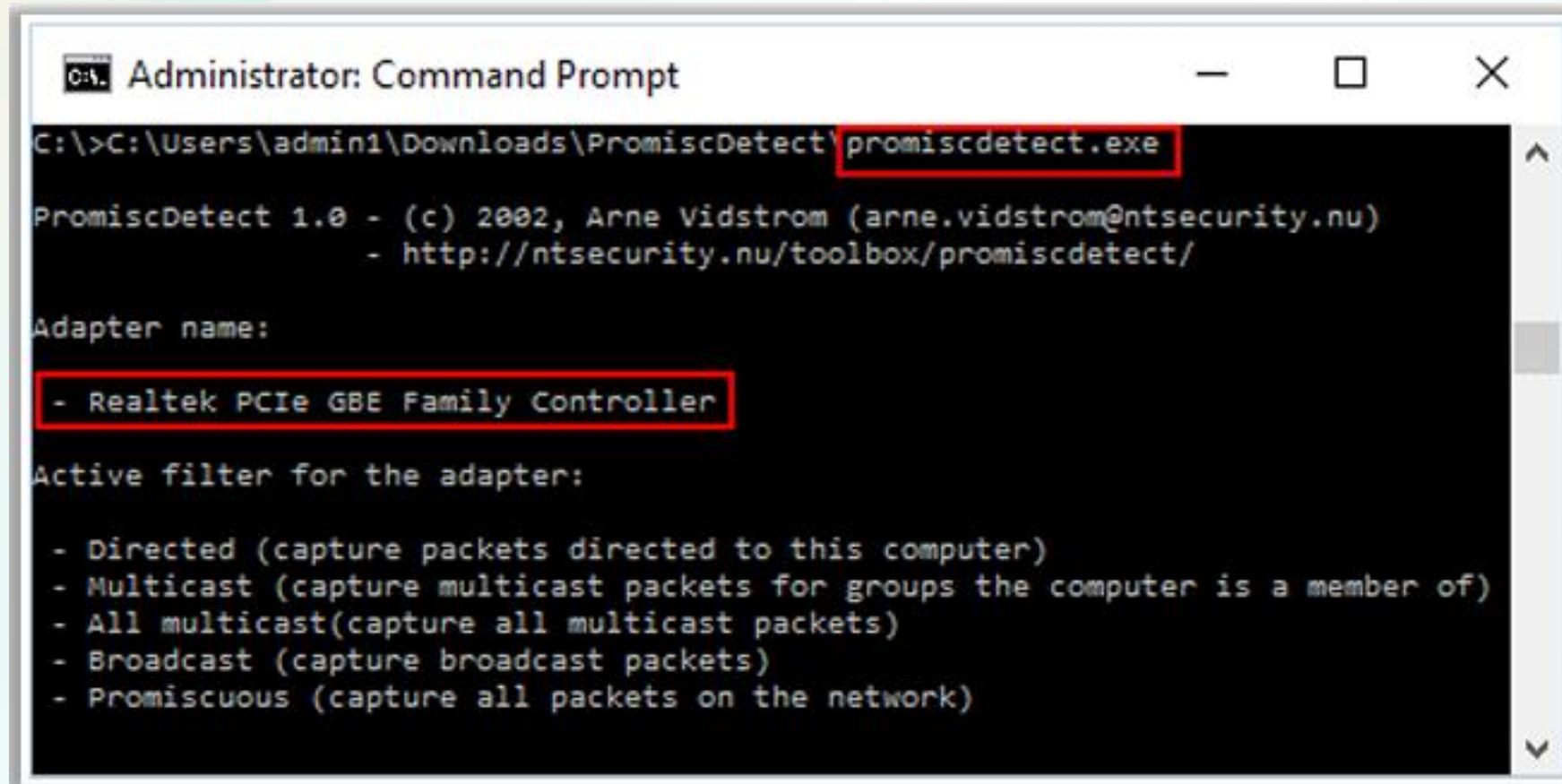
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Broadcom 802.11ac Network Adapter
Physical Address. . . . . : A4-83- -7D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e012:f149:5f48:2a81%14(Preferred)
IPv4 Address. . . . . : 192.168.1.102(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 17 April 2020 11:17:30
Lease Expires . . . . . : 18 April 2020 11:17:30
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 111447
DHCPv6 Client DUID. . . . . : 00-01-00-01-26-1E-80- -3C-7D
DNS Servers . . . . . : 183.83.34.8
                        123.176.37.36
NetBIOS over Tcpip. . . . . : Enabled

C:\Windows\system32>
```

Running the ipconfig /all command





```
Administrator: Command Prompt

C:\>C:\Users\admin1\Downloads\PromiscDetect\promiscdetect.exe

PromiscDetect 1.0 - (c) 2002, Arne Vidstrom (arne.vidstrom@ntsecurity.nu)
- http://ntsecurity.nu/toolbox/promiscdetect/

Adapter name:

- Realtek PCIe GBE Family Controller

Active filter for the adapter:

- Directed (capture packets directed to this computer)
- Multicast (capture multicast packets for groups the computer is a member of)
- All multicast (capture all multicast packets)
- Broadcast (capture broadcast packets)
- Promiscuous (capture all packets on the network)
```

## PromiscDetect

## Collecting Non-volatile Information

-  1 Non-volatile data remain **unchanged** even after the system is shut down or powered off
-  2 **Example:** Emails, word processing documents, spreadsheets and various “deleted” files
-  3 Such data usually resides in **hard drive** (swap file, slack space, unallocated drive space, etc.)
-  4 Other non-volatile data sources include DVDs, **USB thumb** drives, smartphone memory, etc.



# Examining File Systems



- ☐ Run the command **dir /o:d** in command prompt
- ☐ This enables the investigator to examine:
  - The **time** and **date** of the OS installation
  - The service packs, patches, and sub-directories that automatically update themselves often.  
For e.g.: drivers, etc.
- ☐ Give priority to recently dated files

```
Administrator: Command Prompt
03/10/2016 03:30 AM <DIR> migration
03/10/2016 03:32 AM 352,136 FNTCACHE.DAT
03/25/2016 08:09 PM 140,098 perfc009.dat
03/25/2016 08:09 PM 746,532 perfh009.dat
03/25/2016 08:09 PM 883,572 PerfStringBackup.INI
04/06/2016 04:54 PM <DIR> DriverStore
04/13/2016 11:27 AM <DIR> catroot2
04/13/2016 12:33 PM 135,176,864 MRT.exe
04/13/2016 12:33 PM <DIR> MRT
04/14/2016 09:36 AM <DIR> config
04/14/2016 03:06 PM <DIR> drivers
04/14/2016 04:02 PM <DIR> .
04/14/2016 04:02 PM <DIR> ..
04/14/2016 04:02 PM 324 pid.dump
04/14/2016 05:51 PM <DIR> sru
3866 File(s) 1,727,891,022 bytes
116 Dir(s) 63,601,328,128 bytes free

C:\WINDOWS\system32>
```

```
Administrator: Command Prompt

03/10/2016  03:30 AM    <DIR>          migration
03/10/2016  03:32 AM             352,136 FNTCACHE.DAT
03/25/2016  08:09 PM             140,098 perfc009.dat
03/25/2016  08:09 PM             746,532 perfh009.dat
03/25/2016  08:09 PM             883,572 PerfStringBackup.INI
04/06/2016  04:54 PM    <DIR>          DriverStore
04/13/2016  11:27 AM    <DIR>          catroot2
04/13/2016  12:33 PM           135,176,864 MRT.exe
04/13/2016  12:33 PM    <DIR>          MRT
04/14/2016  09:36 AM    <DIR>          config
04/14/2016  03:06 PM    <DIR>          drivers
04/14/2016  04:02 PM    <DIR>          .
04/14/2016  04:02 PM    <DIR>          ..
04/14/2016  04:02 PM             324 pid.dump
04/14/2016  05:51 PM    <DIR>          sru
                3866 File(s)  1,727,891,022 bytes
                116 Dir(s)  63,601,328,128 bytes free
```

Running dir /o:d command



# ESE Database File

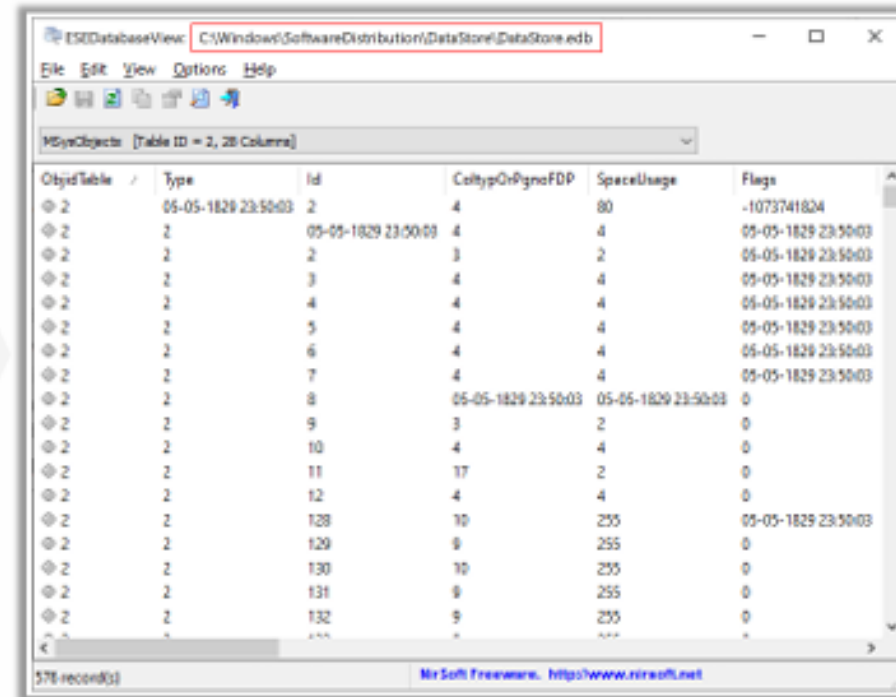
- ❑ Extensible Storage Engine (ESE) is a **data storing technology** used by various Microsoft-managed software such as Active Directory, Windows Mail, Windows Search, and Windows Update Client.
- ❑ This database file is also known as **JET Blue**
- ❑ The file extension of ESE database file is **.edb**. Following are the examples of ESE database files:
  - **contacts.edb** - Stores contacts information in Microsoft live products
  - **WlCalendarStore.edb** - Stores calendar information in Microsoft Windows Live Mail
  - **Mail.MSMessageStore** - Stores messages information in Microsoft Windows Live Mail
  - **WebCacheV24.dat and WebCacheV01.dat** - Stores cache, history, and cookies information in Internet Explorer 10
  - **Mailbox Database.edb and Public Folder Database.edb** - Stores mail data in Microsoft Exchange Server
  - **Windows.edb** - Stores index information (for Windows search) by Windows OS
  - **DataStore.edb** - Stores Windows updates information (Located under C:\windows\SoftwareDistribution\DataStore)
  - **spartan.edb** - Stores the Favorites of Internet Explorer 10/11. (Stored under %LOCALAPPDATA%\Packages\Microsoft.MicrosoftEdge\_8wekyb3d8bbwe\AC\MicrosoftEdge\User\Default\DataStore\Data\nouser1\120712-0049)





# Examining .edb File Using ESEDatabaseView

- ☐ The data stored inside ESE **database files** can be parsed by tools such as **ESEDatabaseView** and **ViewESE**
- ☐ During forensic investigation, the data extracted from these **.edb** files can serve as a potential evidence
- ☐ **ESEDatabaseView** lists all the tables and records found in the selected tables of .edb database file
- ☐ The data extracted from **ESEDatabaseView** can be exported to a HTML file

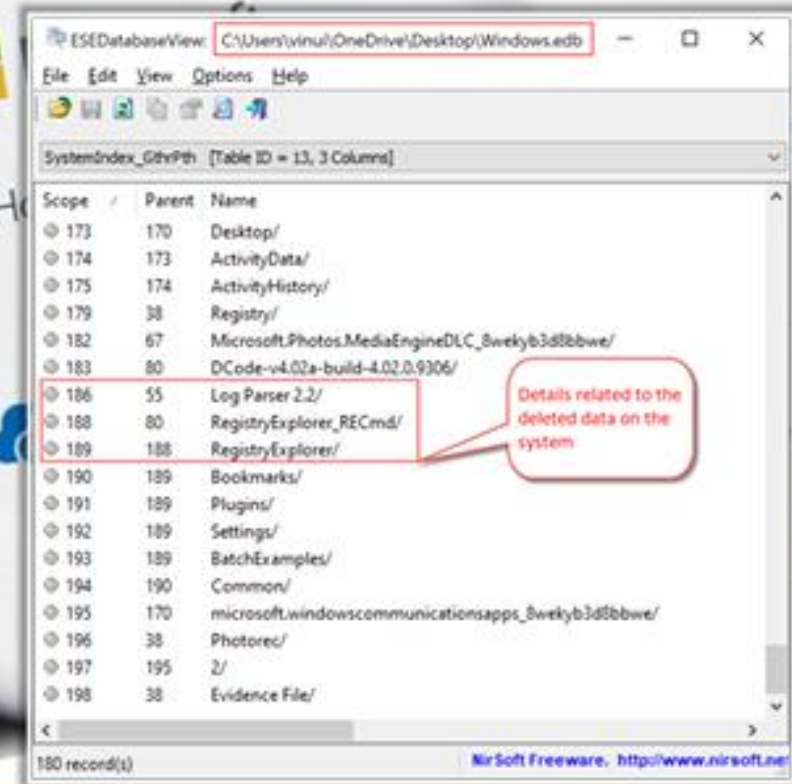


The screenshot shows the ESEDatabaseView application window. The title bar indicates the file path: C:\Windows\SoftwareDistribution\DataStore\DataStore.edb. The menu bar includes File, Edit, View, Options, and Help. Below the menu bar is a toolbar with icons for opening, saving, and other functions. A dropdown menu shows 'MSysObjects [Table ID = 2, 28 Columns]'. The main area displays a table with the following columns: ObjId/TableName, Type, Id, CollType/PrimoFDP, SpaceUsage, and Flags. The table contains 17 rows of data, with the last row showing 578 records. The status bar at the bottom indicates '578 record(s)' and provides a website link: 'Mr Soft Freeware. https://www.mrsoft.net'.

| ObjId/TableName | Type                | Id                  | CollType/PrimoFDP   | SpaceUsage          | Flags               |
|-----------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| 2               | 05-05-1829 23:50:03 | 2                   | 4                   | 80                  | -1073741824         |
| 2               | 2                   | 05-05-1829 23:50:03 | 4                   | 4                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 2                   | 3                   | 2                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 3                   | 4                   | 4                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 4                   | 4                   | 4                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 5                   | 4                   | 4                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 6                   | 4                   | 4                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 7                   | 4                   | 4                   | 05-05-1829 23:50:03 |
| 2               | 2                   | 8                   | 05-05-1829 23:50:03 | 05-05-1829 23:50:03 | 0                   |
| 2               | 2                   | 9                   | 3                   | 2                   | 0                   |
| 2               | 2                   | 10                  | 4                   | 4                   | 0                   |
| 2               | 2                   | 11                  | 17                  | 2                   | 0                   |
| 2               | 2                   | 12                  | 4                   | 4                   | 0                   |
| 2               | 2                   | 128                 | 10                  | 255                 | 05-05-1829 23:50:03 |
| 2               | 2                   | 129                 | 9                   | 255                 | 0                   |
| 2               | 2                   | 130                 | 10                  | 255                 | 0                   |
| 2               | 2                   | 131                 | 9                   | 255                 | 0                   |
| 2               | 2                   | 132                 | 9                   | 255                 | 0                   |

# Windows Search Index Analysis

- ❑ Windows Search Index uses **ESE data storage technology** to store its data
- ❑ It is stored in a file called **Windows.edb**, located in the directory:  
**C:\ProgramData\Microsoft\Search\Data\Applications\Windows**
- ❑ Forensic investigators **parse those files to extract data** pertaining to deleted data, damaged disks, encrypted files, Event bounding, etc., which can be a good source of evidence for investigation
- ❑ In the given screenshot, ESEDatabaseView is used to **parse Windows.edb file** and extract the details of deleted data on the system



# Detecting Externally Connected Devices to the System

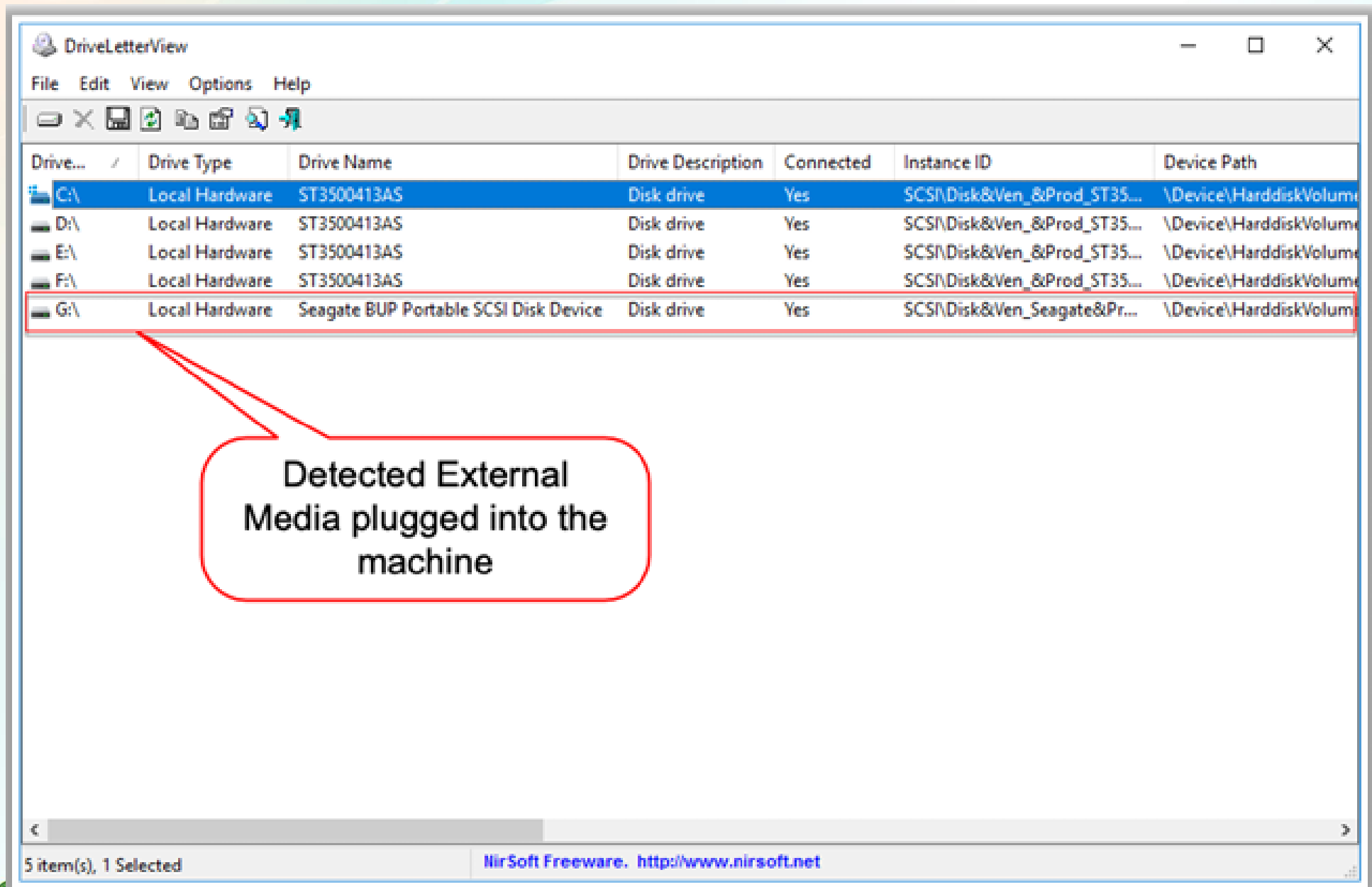


- ❑ Attackers connect external storage media to the system and steal sensitive data or **perform illicit activities**
- ❑ As a part of the forensic investigation, identifying the devices connected to the system helps investigator to determine if any external media is used by the suspect
- ❑ Later, the investigator can get the specific external media from the suspect in a legal manner for further analysis
- ❑ The utility, **DriveLetterView**, lists all the drives on the system even if they are not currently plugged

| Drive... | Drive Type     | Drive Name                            | Drive Description | Connected | Instance ID                   | Device Path             |
|----------|----------------|---------------------------------------|-------------------|-----------|-------------------------------|-------------------------|
| C:       | Local Hardware | ST1500411A5                           | Disk drive        | Yes       | SCSI Disk Drive_&Prod_ST15... | \Device\HarddiskVolume1 |
| D:       | Local Hardware | ST1500411A5                           | Disk drive        | Yes       | SCSI Disk Drive_&Prod_ST15... | \Device\HarddiskVolume2 |
| E:       | Local Hardware | ST1500411A5                           | Disk drive        | Yes       | SCSI Disk Drive_&Prod_ST15... | \Device\HarddiskVolume3 |
| F:       | Local Hardware | ST1500411A5                           | Disk drive        | Yes       | SCSI Disk Drive_&Prod_ST15... | \Device\HarddiskVolume4 |
| G:       | Local Hardware | Seagate BUP Portable SCSI Disk Device | Disk drive        | Yes       | SCSI Disk Drive_SeagateBUP... | \Device\HarddiskVolume5 |

Detected External  
Media plugged into the  
machine

| Dr... | Free Space | Total Size | % Free Space | Cluster Size | Product String | Product Revis... | Vendor... | Serial Number    |
|-------|------------|------------|--------------|--------------|----------------|------------------|-----------|------------------|
| C:    | 27.12 GB   | 63.76 GB   | 42.5%        | 4096         | ST1500411A5    | JC49             | WDA8BPC   |                  |
| D:    | 1.81 GB    | 227.89 GB  | 0.8%         | 4096         | ST1500411A5    | JC49             | WDA8BPC   |                  |
| E:    | 28.53 GB   | 66.36 GB   | 34.4%        | 4096         | ST1500411A5    | JC49             | WDA8BPC   |                  |
| F:    | 27.23 GB   | 67.66 GB   | 27.8%        | 4096         | ST1500411A5    | JC49             | WDA8BPC   |                  |
| G:    | 2.50 TB    | 3.64 TB    | 68.8%        | 262144       | BUP Portable   | 0004             | Seagate   | 00000000P4B100A2 |



# Slack Space

- ❑ Slack space refers to the **portions of a hard drive** that may contain data either from a previously deleted file or unused space by the currently allocated file
- ❑ **Non-contiguous** file allocation leaves more trailing clusters leaving more slack space
- ❑ The data residue in the slack space is retrieved by **reading the complete cluster**
- ❑ **DriveSpy** tool collects all the slack space in an entire partition into a file

## Steps in slack space information collection

Connect to the target computer and select the media

Create bit-level copy of the original media

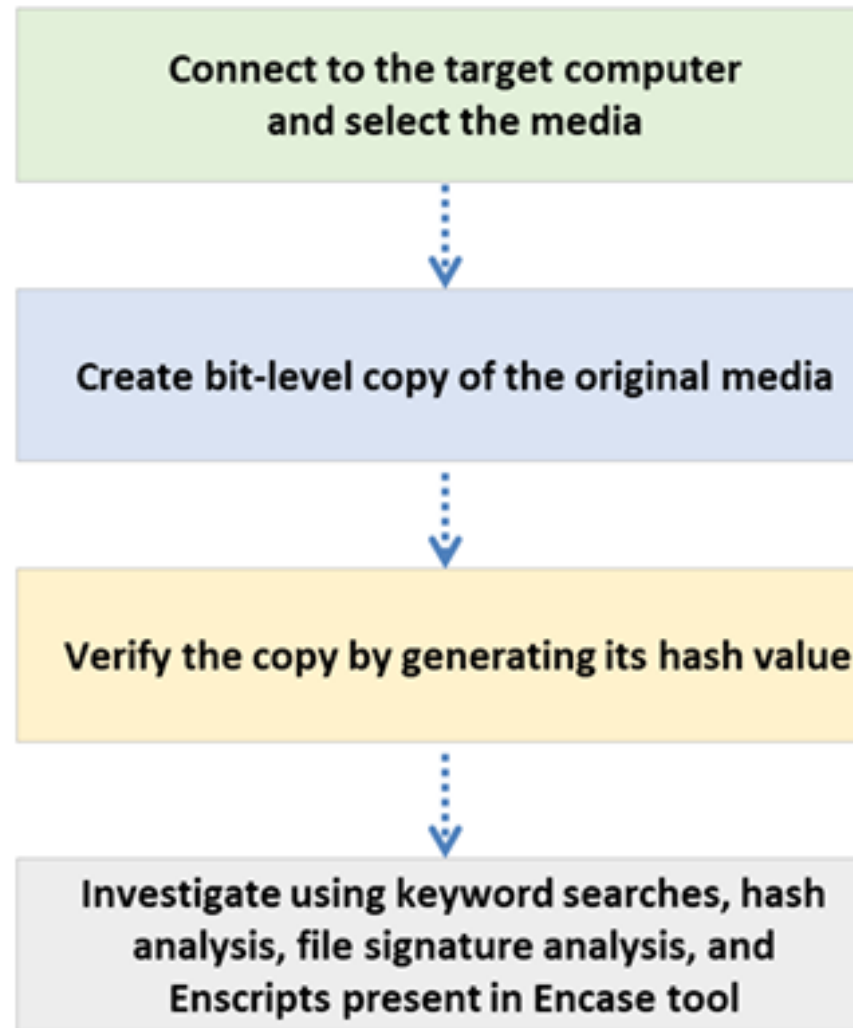
Verify the copy by generating its hash value

Investigate using keyword searches, hash analysis, file signature analysis, and Encrypted present in Encase tool





## Steps in slack space information collection





# Windows Memory Analysis



- ❑ Windows memory analysis involves **acquisition of physical memory or RAM dumps** of the Windows machine
- ❑ Examining these memory dumps help investigators **detect hidden rootkits, find hidden objects**, determine any suspicious process, etc.



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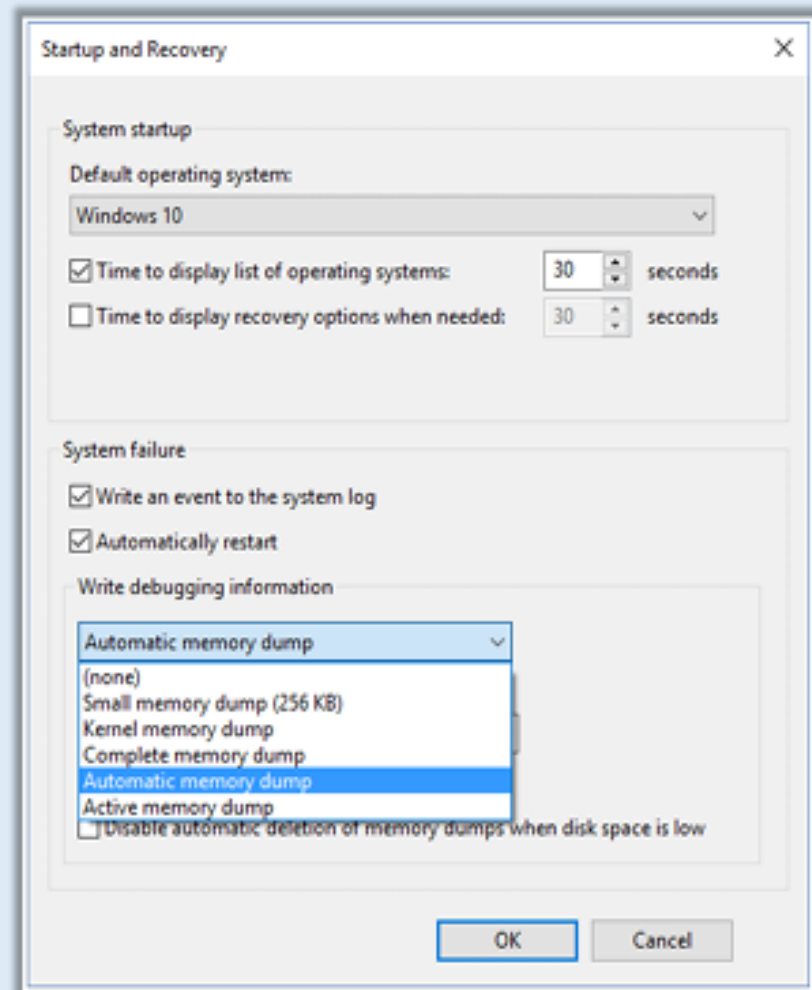


FEU TECH

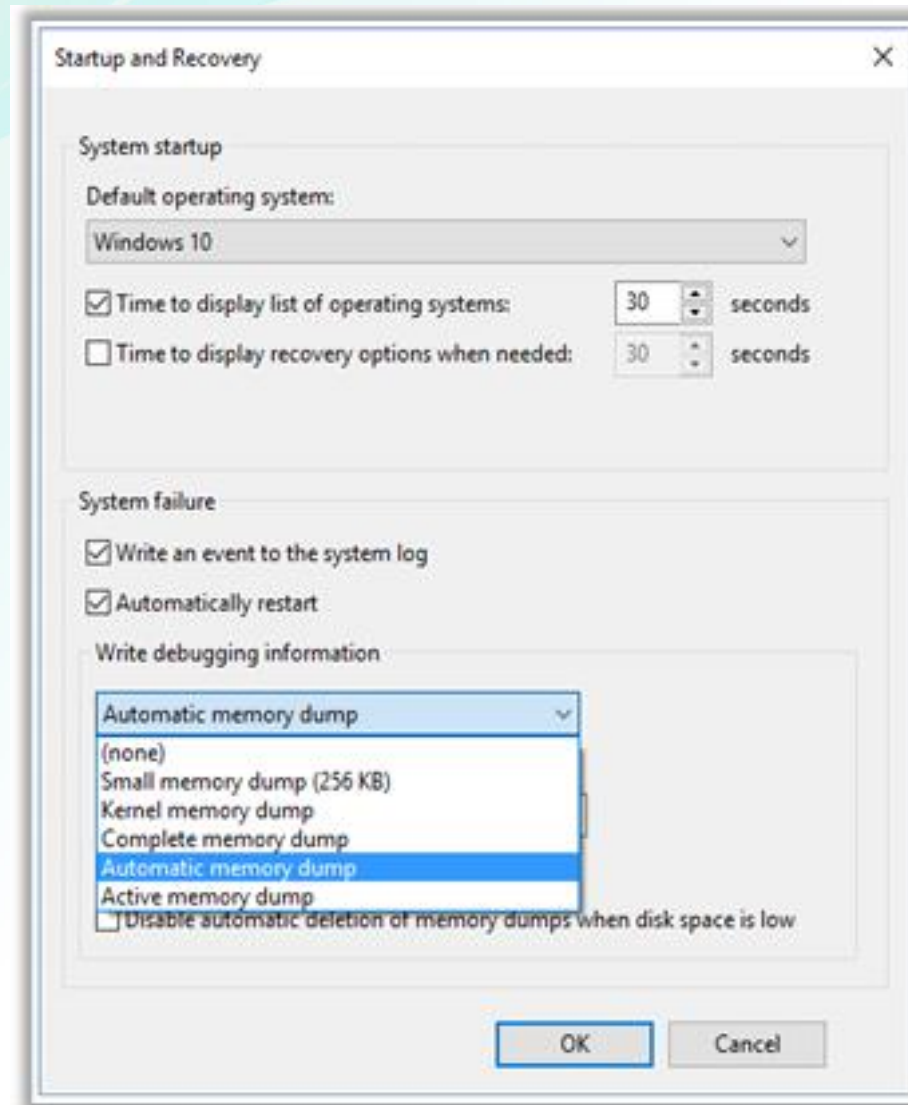
*Technology Driven by Innovation*

# Windows Crash Dump

- ☐ Windows crash dump file contains the contents of computer's memory at the time of a crash
- ☐ It helps in diagnosing and identifying bugs in a program that led to the system crash
- ☐ You can check the memory dump information using **DumpChk** utility
- ☐ In Windows 10, the OS creates the following memory dumps:
  - Automatic memory dump
  - Complete memory dump
  - Kernel memory dump
  - Small memory dump
- ☐ **Examining the crash dumps** can sometimes help a forensic investigator in finding out if the crash is caused due to an internal error or by a remote attacker, who was successful in exploiting a bug in the OS, or a third-party application installed on the OS







## Selecting Automatic memory dump in Startup and Recovery Settings



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# Collecting Process Memory

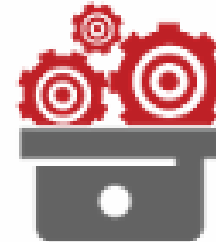
Collect the contents of process memory available in a **RAM dump file**

**Process Dumper** (**pd.exe**) dumps the entire process space along with the additional metadata and the process environment to the console; it redirects the output to a file or a socket

**Userdump.exe** dumps any process without attaching a debugger and without terminating the process once the dump has been completed

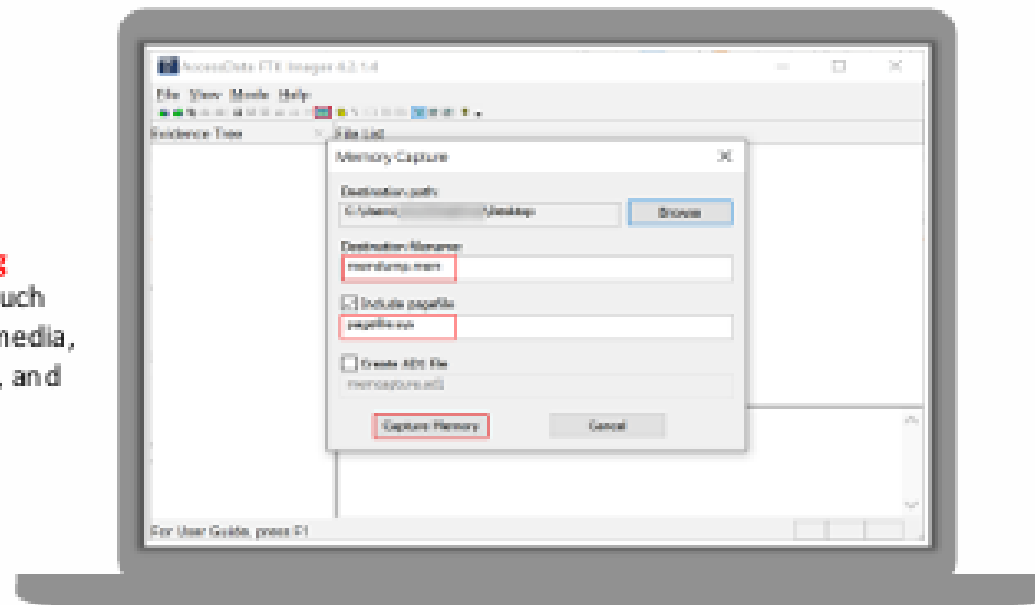
Another method of dumping a process is to use **adplus.vbs** script

Once done with the dumping process, use **debugging tools** to analyze the dump files



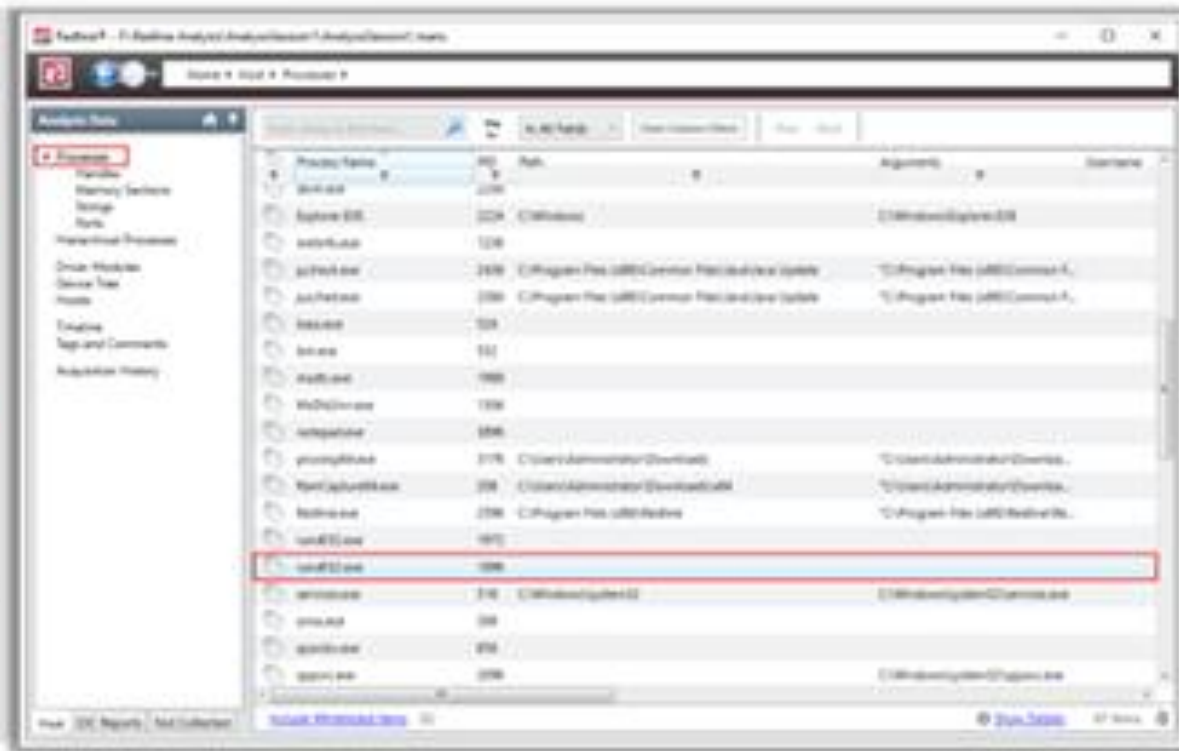
# Random Access Memory (RAM) Acquisition

- 01 Examining **volatile memory** is as important as non-volatile memory
- 02 From forensics point of view, **examining RAM dumps** provides system artifacts such as running services, accessed files and media, system processes, network information, and malware activity
- 03 During **live acquisition**, investigators use tools such as **Belkasoft RAM Capturer** and **AccessData FTK Imager** to perform RAM dumps



## Memory Forensics: Malware Analysis Using Redline

- ❑ Redline is a security tool to identify malicious activities through memory and helps forensic investigators to establish the **timeline and scope of an incident**
- ❑ Analyze the RAM dump using Redline by loading it from '**Analyze Data**' section
- ❑ Under '**Analysis Data**' tab, you can find all the processes running on the system when the RAM dump was acquired



<http://www.finnovet.com>

Redline® - F:\Redline Analysis\AnalysisSession1\AnalysisSession1.mans

Home ▶ Host ▶ Processes ▶

Analysis Data

Processes

Handles  
Memory Sections  
Strings  
Ports  
Hierarchical Processes  
Driver Modules  
Device Tree  
Hooks  
Timeline  
Tags and Comments  
Acquisition History

Enter string to find here...

Reg Ex In All Fields Clear Column Filters Prev Next

| Process Name     | PID  | Path                                                 | Arguments                             | Username |
|------------------|------|------------------------------------------------------|---------------------------------------|----------|
| dwm.exe          | 2200 |                                                      |                                       |          |
| Explorer.EXE     | 2224 | C:\Windows                                           | C:\Windows\Explorer.EXE               |          |
| inetinfo.exe     | 1236 |                                                      |                                       |          |
| jucheck.exe      | 2436 | C:\Program Files (x86)\Common Files\Java\Java Update | "C:\Program Files (x86)\Common F...   |          |
| jusched.exe      | 2384 | C:\Program Files (x86)\Common Files\Java\Java Update | "C:\Program Files (x86)\Common F...   |          |
| lsass.exe        | 524  |                                                      |                                       |          |
| lsmon.exe        | 532  |                                                      |                                       |          |
| msdtc.exe        | 1988 |                                                      |                                       |          |
| MsDtsSrvr.exe    | 1304 |                                                      |                                       |          |
| notepad.exe      | 3896 |                                                      |                                       |          |
| procexp64.exe    | 3176 | C:\Users\Administrator\Downloads                     | "C:\Users\Administrator\Downloa...    |          |
| RamCapture64.exe | 208  | C:\Users\Administrator\Downloads\x64                 | "C:\Users\Administrator\Downloa...    |          |
| Redline.exe      | 2596 | C:\Program Files (x86)\Redline                       | "C:\Program Files (x86)\Redline\Re... |          |
| rundll32.exe     | 1972 |                                                      |                                       |          |
| rundll32.exe     | 1896 |                                                      |                                       |          |
| services.exe     | 516  | C:\Windows\system32                                  | C:\Windows\system32\services.exe      |          |
| smss.exe         | 268  |                                                      |                                       |          |
| spoolsv.exe      | 856  |                                                      |                                       |          |
| sppsvc.exe       | 2096 |                                                      | C:\Windows\system32\sppsvc.exe        |          |

Host IOC Reports Not Collected

Include Whitelisted Items

Show Details 67 Items

Identifying suspicious process

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Redline® - F:\Redline Analysis\AnalysisSession1\AnalysisSession1.mans

Home ▶ Host ▶ Processes ▶ Ports

Analysis Data

- Processes
  - Handles
  - Memory Sections
  - Strings
  - Ports
- Hierarchical Processes
- Driver Modules
- Device Tree
- Hooks
- Timeline
- Tags and Comments
- Acquisition History

Filters

Enter string to find here...

Reg Ex In All Fields Clear Column Filters Prev Next

| Process Name | PID  | Path              | State       | Created     | Local IP Address   | Local Port | Remote IP Address | Remote Port | Protocol |
|--------------|------|-------------------|-------------|-------------|--------------------|------------|-------------------|-------------|----------|
| jusched.exe  | 2384 | C:\Program Fil... | CLOSE_WAIT  |             | 192.168.1.100      | 492...     | 23.59.188.113     | 80          | TCP      |
| rundll32.exe | 1972 |                   | CLOSED      |             | 172.20.20.9        | 530...     | 172.20.20.21      | 4444        | TCP      |
| chrome.exe   | 2268 | C:\Program Fil... | LISTENING   | 2020-03-... | 00:00:00:00:00:... | 5353       | *:*               | 0           | UDP      |
| chrome.exe   | 2268 | C:\Program Fil... | LISTENING   | 2020-03-... | 192.168.1.100      | 545...     | *:*               | 0           | UDP      |
| svchost.exe  | 2012 |                   | LISTENING   |             | 00:00:00:00:00:... | 3389       |                   | 0           | TCP      |
| svchost.exe  | 1244 |                   | LISTENING   | 2020-03-... | 0.0.0.0            | 0          | *:*               | 0           | UDP      |
| svchost.exe  | 1244 |                   | LISTENING   |             | 0.0.0.0            | 491...     |                   | 0           | TCP      |
| rundll32.exe | 1896 |                   | CLOSED      |             | 172.20.20.9        | 530...     | 172.20.20.21      | 4444        | TCP      |
| rundll32.exe | 1896 |                   | ESTABLISHED |             | 172.20.20.9        | 492...     | 172.20.20.21      | 4444        | TCP      |
| svchost.exe  | 792  |                   | LISTENING   |             | 0.0.0.0            | 491...     |                   | 0           | TCP      |
| svchost.exe  | 976  |                   | LISTENING   | 2020-03-... | 00:00:00:00:00:... | 0          | *:*               | 0           | UDP      |
| svchost.exe  | 976  |                   | LISTENING   | 2020-03-... | 00:00:00:00:00:... | 5355       | *:*               | 0           | UDP      |
| wininit.exe  | 420  |                   | LISTENING   |             | 00:00:00:00:00:... | 491...     |                   | 0           | TCP      |
| services.exe | 516  | C:\Windows\s...   | LISTENING   |             | 0.0.0.0            | 491...     |                   | 0           | TCP      |
| lsass.exe    | 524  |                   | LISTENING   |             | 00:00:00:00:00:... | 491...     |                   | 0           | TCP      |
| svchost.exe  | 704  |                   | LISTENING   |             | 0.0.0.0            | 135        |                   | 0           | TCP      |
| svchost.exe  | 836  | C:\Windows\s...   | LISTENING   | 2020-03-... | 0.0.0.0            | 0          | *:*               | 0           | UDP      |
| svchost.exe  | 836  | C:\Windows\s...   | LISTENING   | 2020-03-... | 00:00:00:00:00:... | 4500       | *:*               | 0           | UDP      |
| svchost.exe  | 826  | C:\Windows\s...   | LISTENING   |             | 00:00:00:00:00:... | 401        |                   | 0           | TCP      |

Host IOC Reports Not Collected

Show Details 39 Items

Using Redline for malware analysis

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# Windows Registry Analysis



The Windows registry is a **hierarchical database** that contains low-level settings for the Microsoft Windows OS and for applications that use the registry



Investigating the data present in the registry help forensic investigators obtain information on software installed and **hardware driver's configuration settings**, track suspicious user activity, etc.



This information help investigators build **timeline analysis** of the incident during forensic investigation



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# Windows Registry

- ❑ Every action performed by the user on the machine is **recorded in the Windows Registry**; Hence, it is a good source of evidence during forensic investigation
- ❑ With respect to data persistence, Windows Registry hives are divided into:

## Non-volatile:

HKEY\_LOCAL\_MACHINE  
HKEY\_USERS

## Volatile:

HKEY\_CLASSES\_ROOT  
HKEY\_CURRENT\_USER  
HKEY\_CURRENT\_CONFIG

- ❑ The volatile hives are captured during **live analysis** of the system while the non-volatile hives are stored on the hard drive



# Windows Registry (Cont'd)

Hives in the Windows registry play a critical role in the functioning of the system:

HKEY\_USERS



It contains all the **actively loaded user profiles** for that system

HKEY\_CLASSES\_ROOT



This hive contains **configuration information** related to the applications used for opening various files on the system

HKEY\_CURRENT\_CONFIG



This hive contains the **hardware profile** the system uses at startup

HKEY\_LOCAL\_MACHINE



This hive contains a vast array of configuration information for the system, including hardware settings and software settings

HKEY\_CURRENT\_USER



It is the active, loaded user profile for the **currently logged-on user**



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# Registry Structure within a Hive File

- ❑ Various components of the registry called “**cells**” have a specific structure and contain specific type of information

## Types of cells:

### Key cell

It contains **Registry key information** and includes offsets to other cells as well as the **LastWrite** time for the key

### Value cell

It holds a **value** and its data



### Subkey list cell

It is made up of a series of indexes pointing to **key cells**, these all are sub keys to the parent key cell

### Value list cell

It is made up of a series of indexes pointing to value cells, these all are values of a **common key cell**

### Security descriptor cell

It contains **security descriptor** information for a key cell

# Windows Registry: Forensic Analysis



- ❑ Forensic analysis of Windows registry helps the investigator to **extract forensic artifacts** such as user accounts, recently accessed files, USB activity, last run programs, and installed applications
- ❑ The forensic investigator should analyze the Windows registry in two methods:

## Static Analysis

- ❑ The investigator examines the registry files stored on the captured evidence file. These files are located in the **C:\Windows\System32\config** folder.

## Live Analysis

- ❑ The investigator can use **built-in registry editor** to examine registry and also use tools like FTK Imager to capture registry files from live system for analysis



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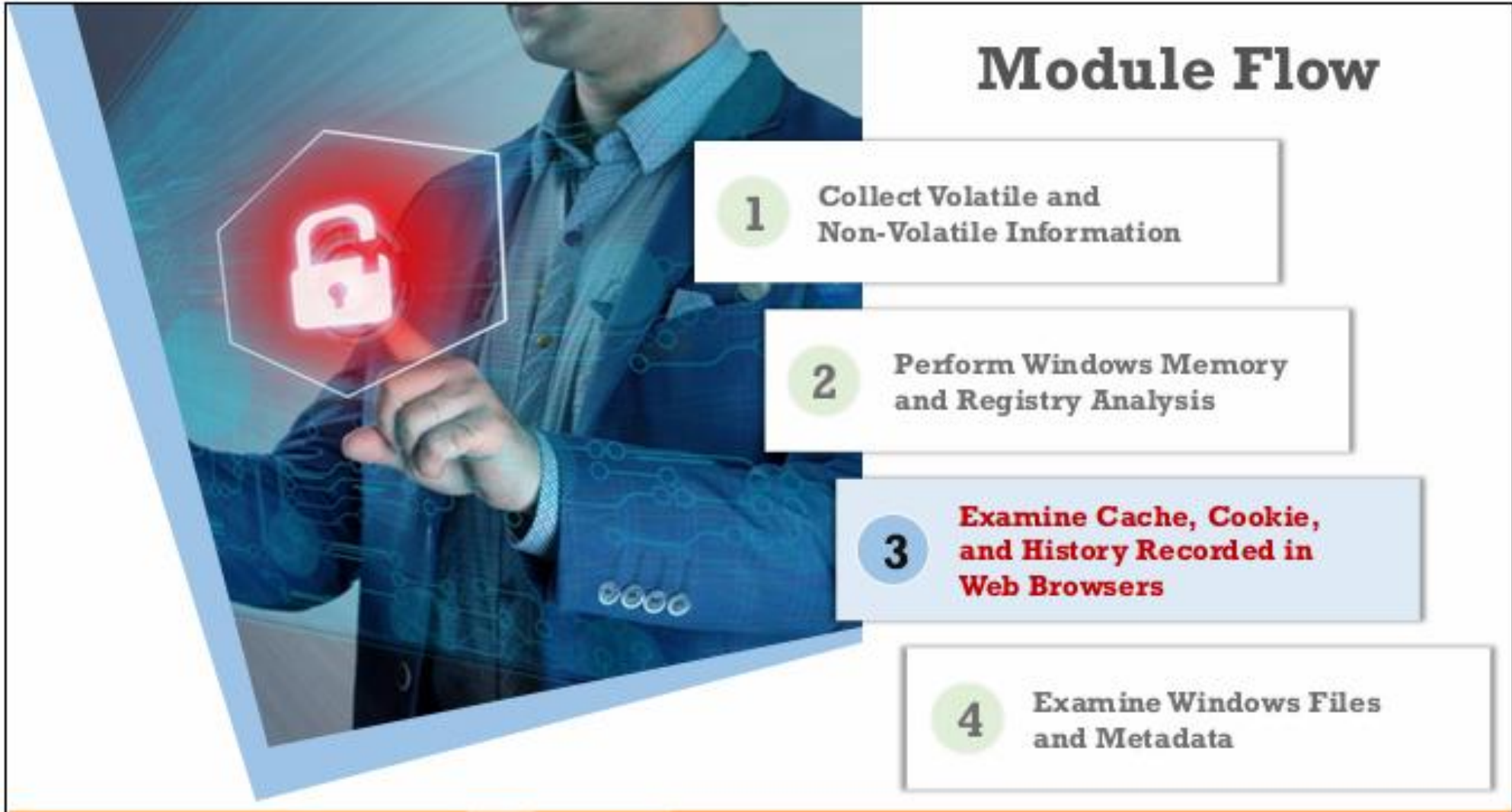
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# Cache, Cookie, and History Analysis

Examining web browsers such as Microsoft Edge, Google Chrome, Mozilla Firefox, etc., provide crucial evidentiary data such as **web history**, **cookie** and **cache information** pertaining to the user's browsing activity



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## Cache, Cookie, and History Analysis: Google Chrome

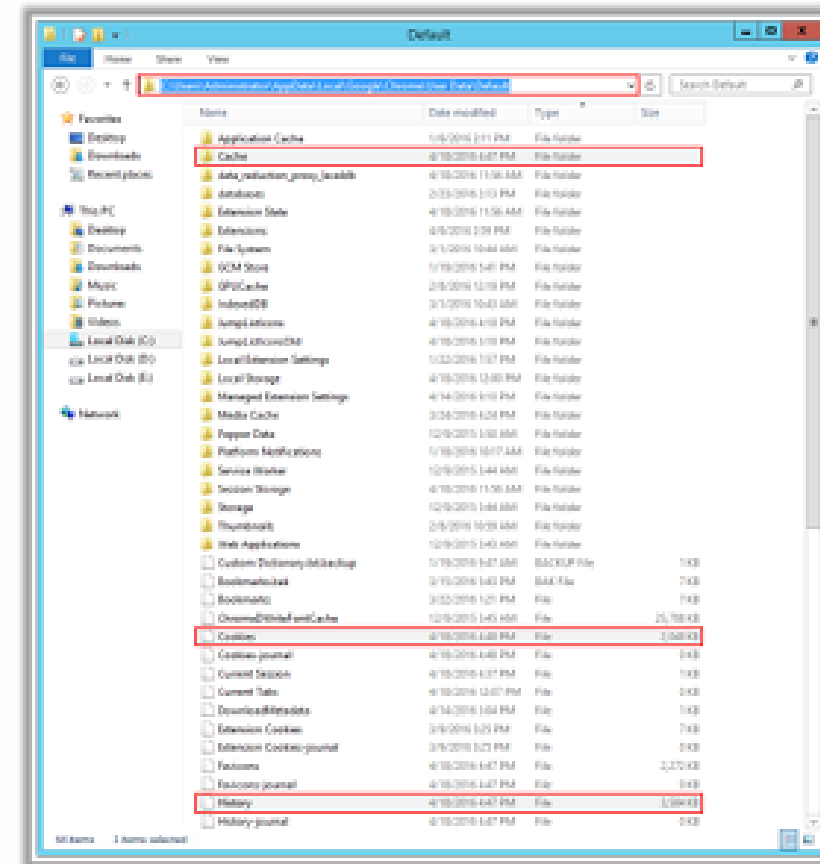
**Google Chrome - Cache, cookies, and history are stored in the following system locations:**

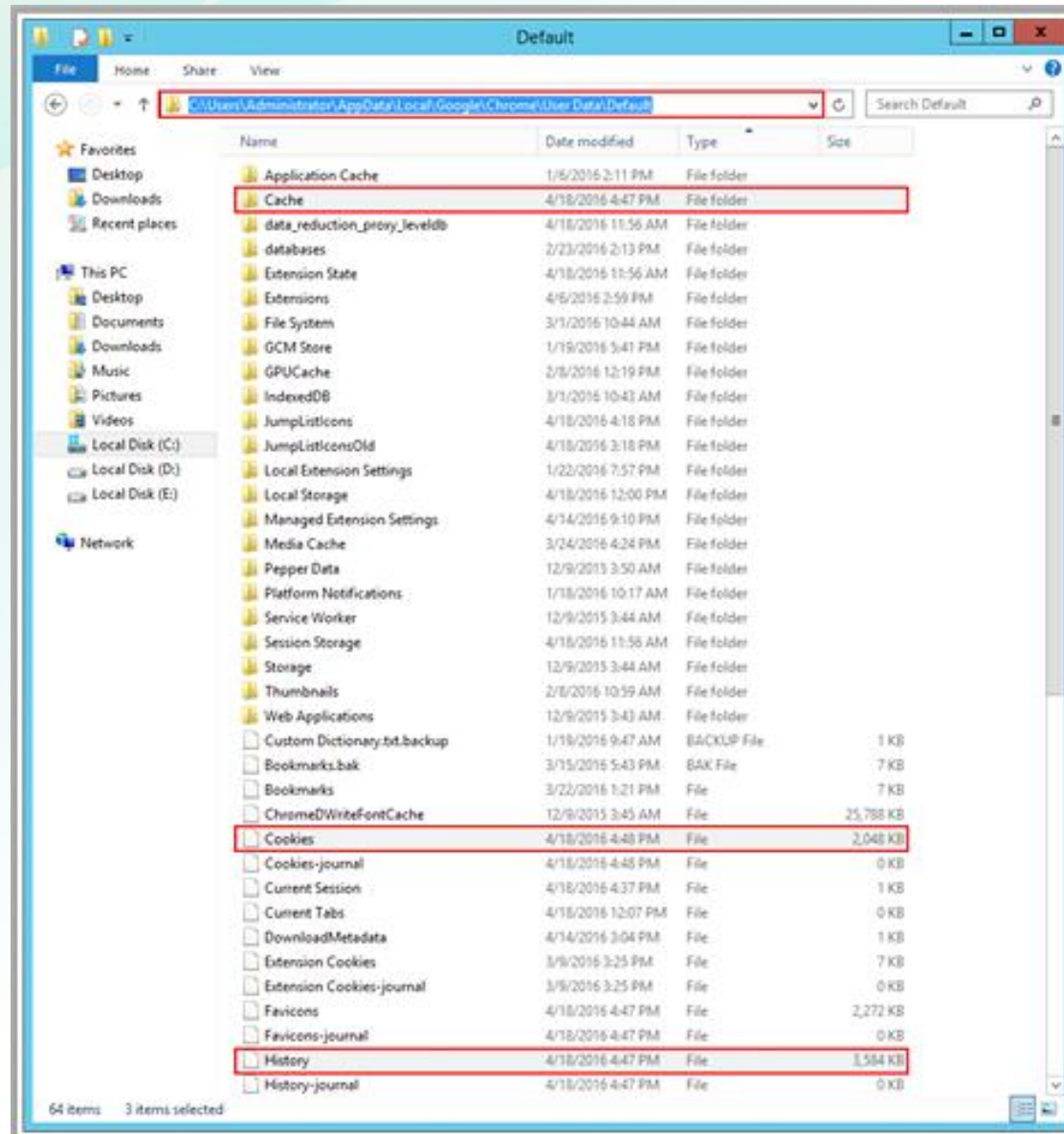
### History and Cookies Location:

C:\Users\{user}\AppData\Local\Google\Chrome\User Data\Default

**Cache Location:**

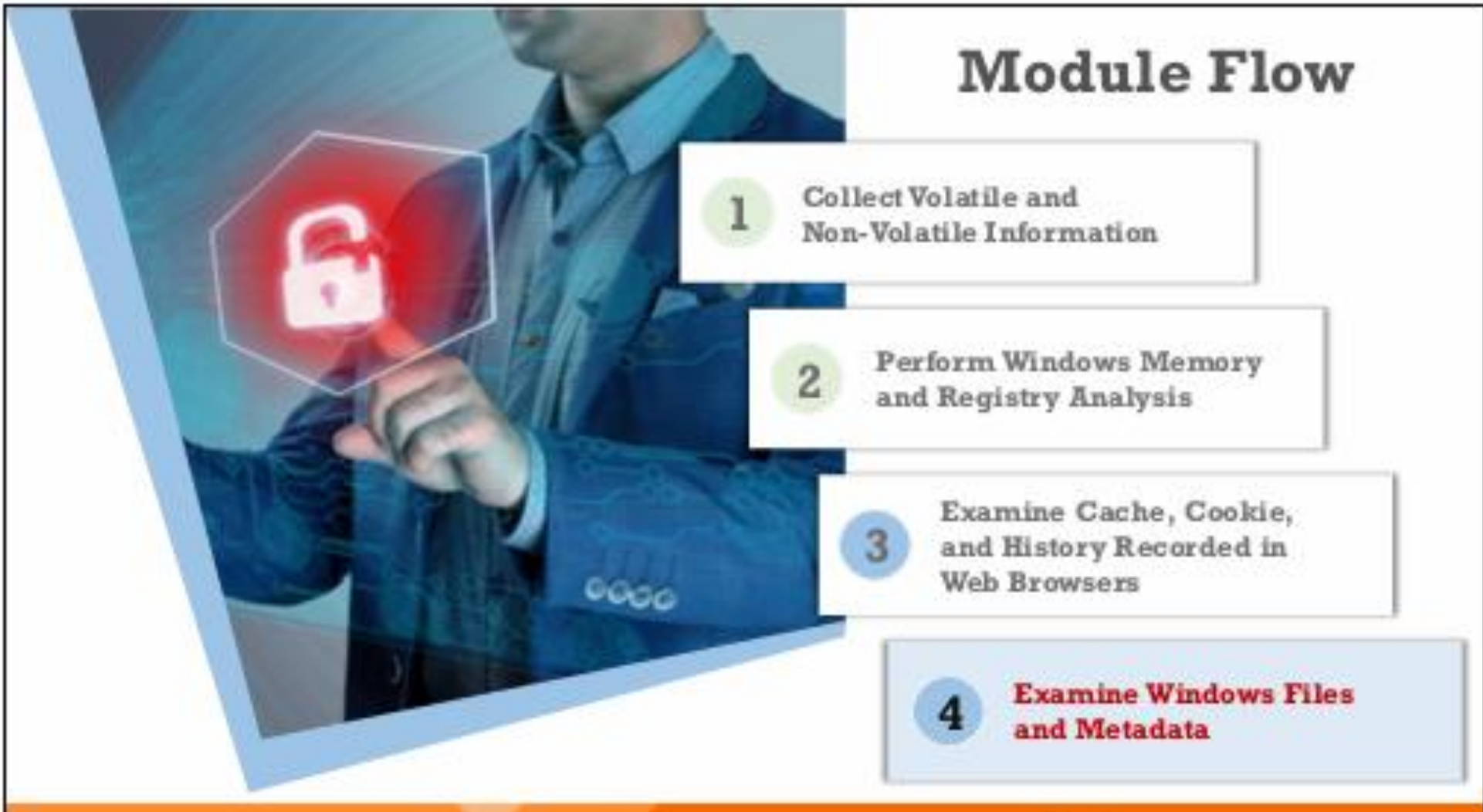
C:\Users\{user}\AppData\Local\Google\Chrome\User Data\Default\Cache





Chrome history, cookies and cache location

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# Windows File Analysis



- ❑ Forensic examination of restore point log files and prefetch files provide information such as **MAC timestamps, file name, file size, number of times the application has been run, process name**, etc., related to the installed/uninstalled applications



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## System Restore Points (Rp.log Files)



Rp.log is the **restore point log** file located within the restore point (RPxx) directory



It includes value indicating the **type of the restore point**; a descriptive name for the restore point creation event, and the 64-bit FILETIME object indicating when the restore point was created



System restore points are created when applications and unsigned drivers are **installed**, when an auto update installation and a restore operation are performed



Description of the event that caused the restore point creation is written to the rp.log file, and this log file helps the **investigator to notice the date** when the application was installed or removed

# Prefetch Files



When a user installs an application, runs it, and deletes it, traces of that application can be found in the **Prefetch directory**



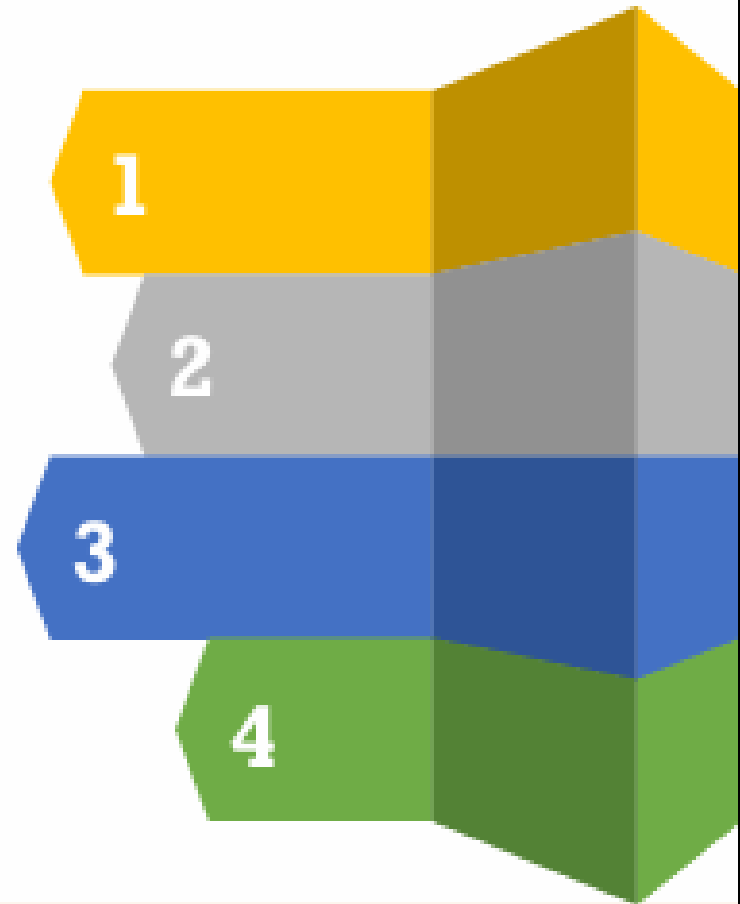
DWORD value at the **offset 144** within the file corresponds to the number of times the application is launched



DWORD value at the **offset 120** within the file corresponds to the last time of the application run, this value is stored in **UTC format**



Information from **.pf file** can be correlated with the registry or Event Log information to determine who was **logged on to the system**, who was running which applications, etc.



# Image Files



The **metadata** present in a JPEG image file depends largely on the application that created or modified it



For e.g., digital cameras embed Exchangeable Image File Format (EXIF) information in images, which can include the model and manufacturer of the camera, and even store thumbnails or audio information



You can use tools such as **Exiv2**, **IrfanView**, and the **Image::MetaData::JPEG** Perl module to view, retrieve, and in some cases modify the metadata embedded in JPEG image files



Tools such as **ExifReader**, **EXIF Library**, and **ExifTool** display **EXIF** data found in a JPEG image



# Metadata Investigation



- ☐ In computer forensics, metadata obtained from the databases, image files, word files, web browsers, etc., contains **evidentiary data** of forensic value
- ☐ Metadata includes **file name**, **file size**, **MAC timestamps**, **GPS data**, etc.



# Understanding Metadata

- ☐ **Metadata** is data about data. It describes various characteristics of data, including when and by whom it was created, accessed, or modified
- ☐ Because it is not normally seen, users can inadvertently **share confidential information** when sending or providing files in electronic form

## Examples of metadata:

- |                        |                                                |
|------------------------|------------------------------------------------|
| • Organization name    | • Document versions                            |
| • Author name          | • Template information                         |
| • Computer name        | • Personalized views                           |
| • Network name         | • Non-visible portions of embedded OLE objects |
| • Hidden text or cells |                                                |

- ☐ The investigator can use tools such as **Metadata Assistant**, **Paraben P2 Commander**, and **Metashield Analyzer** to analyze metadata



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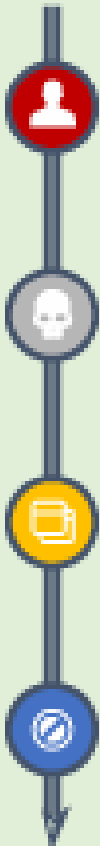
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# Metadata in Different File Systems



- ❑ The most commonly known metadata about files on Windows systems are the files' **MAC times**; MAC stands for **modified**, **accessed**, and **created**
- ❑ The MAC times are **time stamps** that refer to the time at which the file was last modified, last accessed, and originally created
- ❑ **MAC times** are managed by the OS depending on the file system used
  - On the **FAT file system**, times are stored based on the **local time** of the computer system
  - **NTFS file system** stores MAC times in **Coordinated Universal Time (UTC)** format
- ❑ Investigate the way the **timestamps are displayed**, based on various move and copy actions



# Metadata in Different File Systems (Cont'd)

How time stamps are displayed and changed in the FAT 16 and NTFS file systems is shown below

## FAT 16 file system

- ☐ Copy myfile.txt from C:\ to C:\subdir on the same file system (FAT 16)
  - **Myfile.txt** retains the same modification date, but the creation date is updated to the current date and time
- ☐ Move myfile.txt from C:\ to C:\subdir on the same file system (FAT 16)
  - Myfile.txt retains the same modification and creation dates
- ☐ Copy myfile.txt from a FAT16 partition to an NTFS partition
  - Myfile.txt retains the same modification date, but the creation date is updated to the current date and time
- ☐ Move myfile.txt from a FAT16 partition to an NTFS partition
  - Myfile.txt retains the same modification and creation dates

## NTFS file system

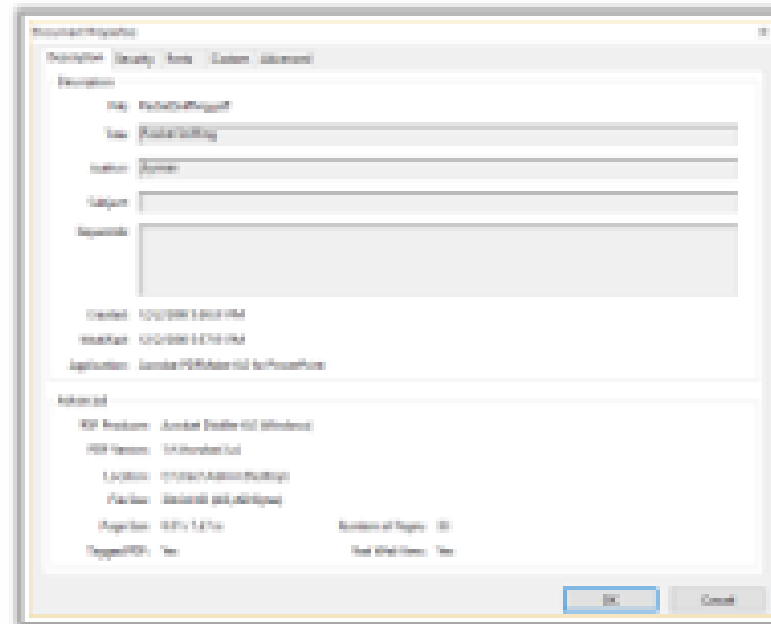
- ☐ Copy myfile.txt from C:\ to C:\subdir on the same file system (NTFS)
  - Myfile.txt retains the same modification date, but the creation date is updated to the current date and time
- ☐ Move myfile.txt from C:\ to C:\subdir on the same file system (NTFS)
  - Myfile.txt retains the same modification and creation dates

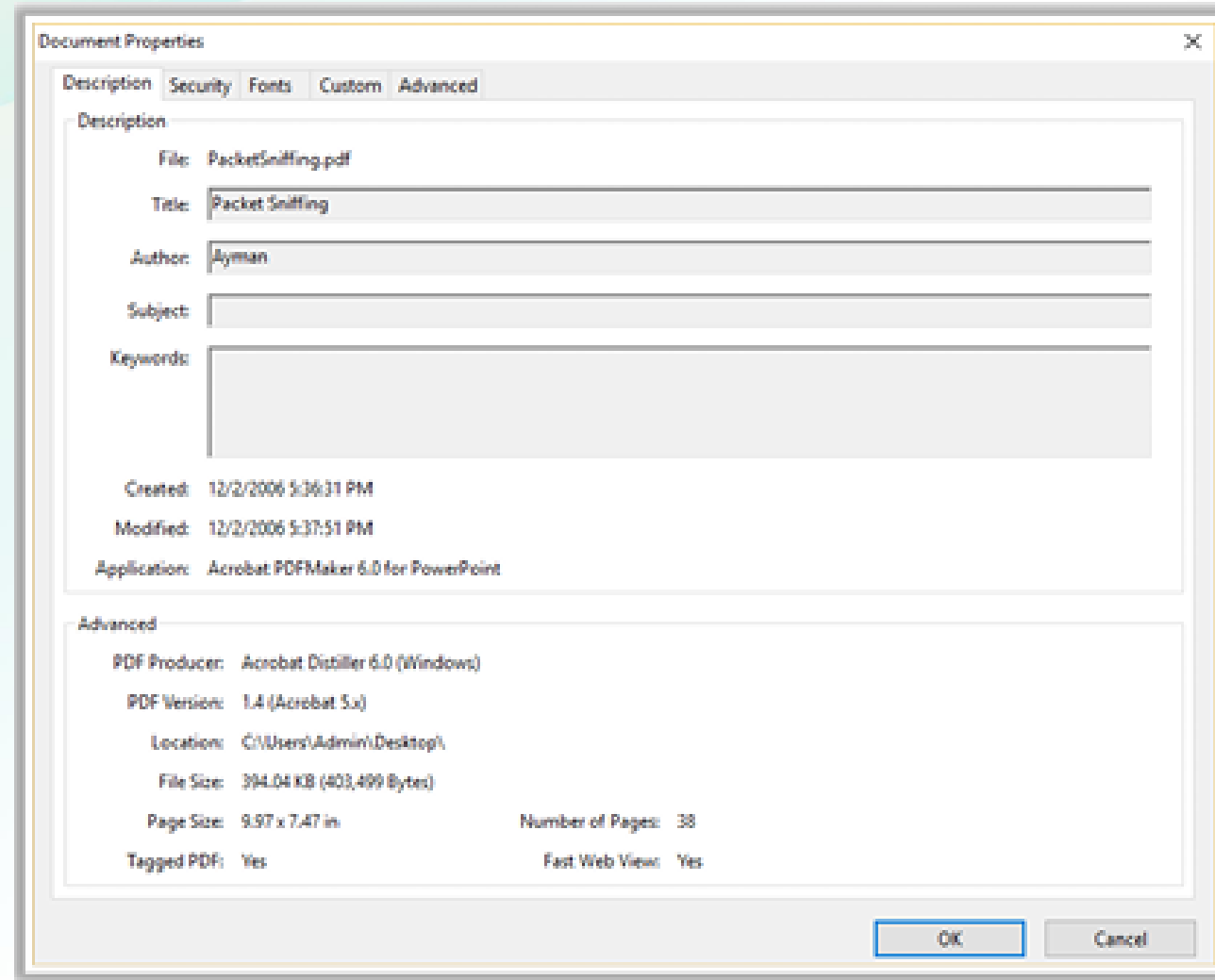


# Metadata in PDF Files

- ❑ Portable document format (PDF) files can also contain **metadata** such as name of the author, the date when file was created, and the application used to create the PDF file
- ❑ Often, the metadata can show that the PDF file was created on a Mac or that the PDF file was created by converting a Word document to PDF format
- ❑ You can use the Perl scripts **pdfmeta.pl** and **pdfdimp.pl** to extract metadata from PDF files

- To view PDF metadata, open it with Adobe Acrobat Reader
- In the **File** menu, click **Properties...**



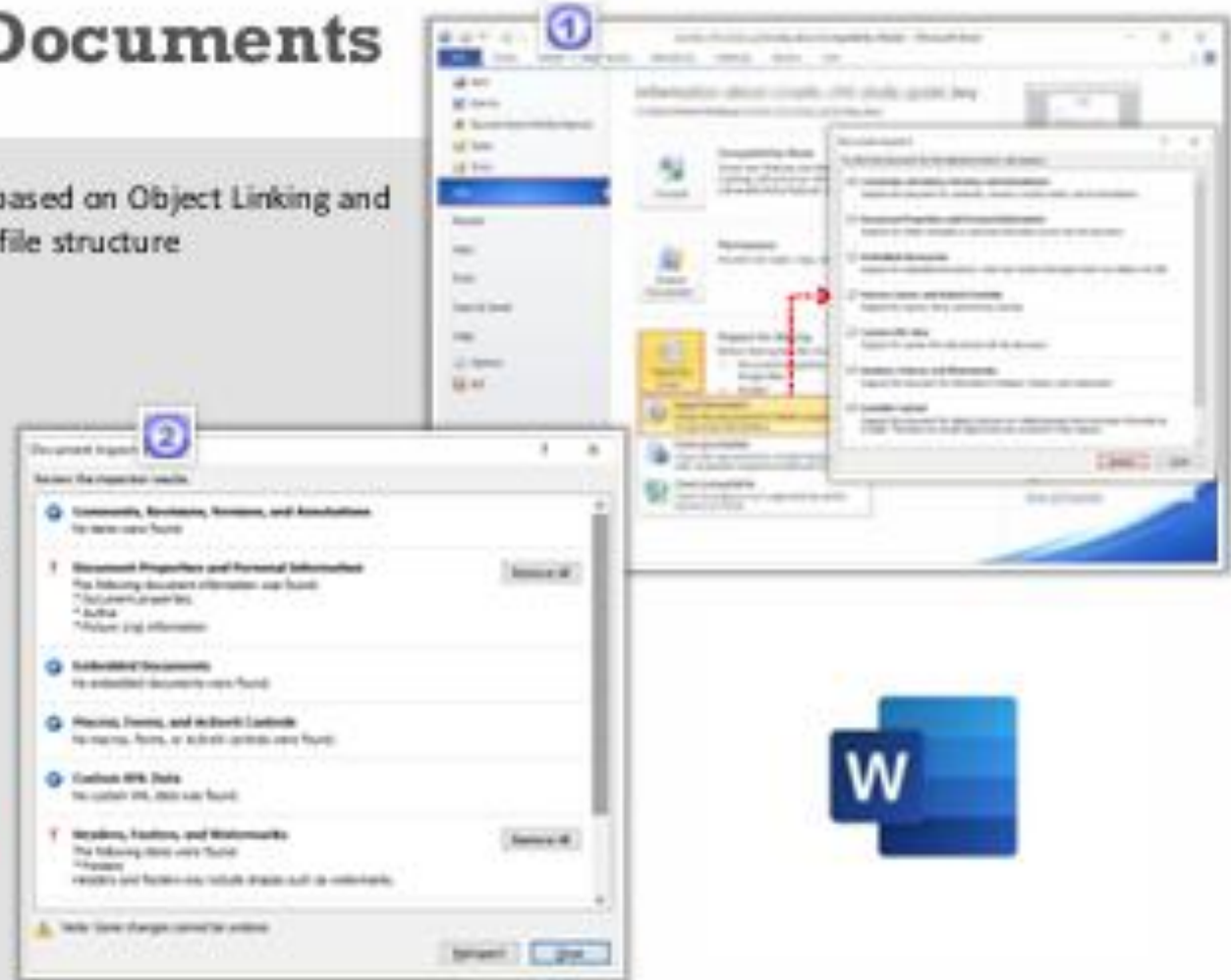


**Document Properties window**



# Metadata in Word Documents

- ☐ Word documents are compound documents, based on Object Linking and Embedding (OLE) technology that defines the file structure
- ☐ Word documents can maintain not only past revisions but also a list of up to the last 10 authors to edit the file
- ☐ You can use the Perl scripts **wmd.pl** and **oledmp.pl** to list the OLE streams embedded in a Word document
- ☐ To view metadata in Word 2010, click on the **File** tab → **Info** option
- ☐ Click **Check for Issues** → **Inspect Document**
- ☐ Select the content to view and click the **Inspect** button



## Metadata Analysis Tool: Metashield Analyzer

- ❑ Metashield Analyzer is an **online service** that allows investigators to **analyze the metadata** contained in files.



# Module Summary

- 1 This module has discussed the collection of volatile and non-volatile information
- 2 It also discussed the analysis of Windows memory and Registry
- 3 Further, it explained in detail the process of examining the cache, cookie, and history recorded in web browsers
- 4 Finally, this module ended with a detailed discussion on examining Windows files and metadata
- 5 In the next module, we will discuss Linux and Mac forensics in detail



# Linux and Mac Forensics

## SUBTOPIC 2





## Module Objectives

- 1 Understanding the Volatile and Non-Volatile Data in Linux
- 2 Understanding the Filesystem Images Analysis using The Sleuth Kit
- 3 Understanding the Memory Forensics using Volatility and PhotoRec
- 4 Understanding the Mac Forensics



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# Module Flow

**Understand Volatile  
and Non-Volatile  
Data in Linux**

01

**Analyze Filesystem  
Images Using The  
Sleuth Kit**

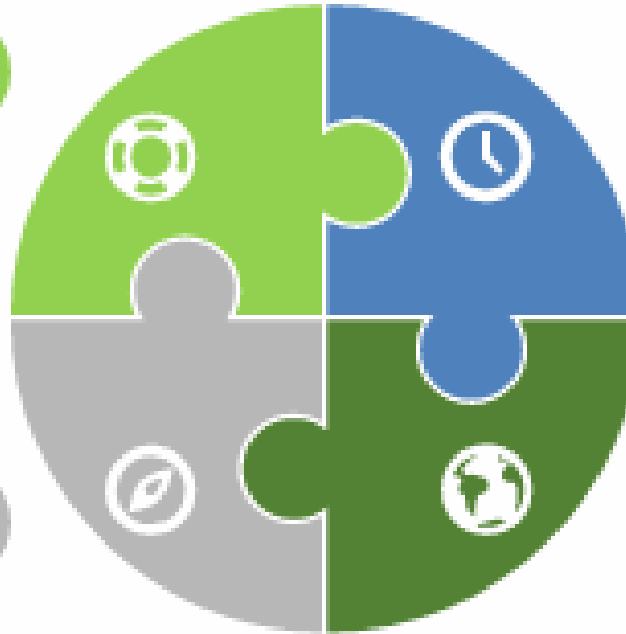
02

**Understand Mac  
Forensics**

04

**Demonstrate Memory  
Forensics**

03



# Introduction to Linux Forensics



Linux is an **open-source OS** that is used extensively for enterprise servers and employee desktop platforms in several corporate organizations



Cybercrimes have been increasing in such **business environments**, and extracting artifacts from Linux systems is a challenging task.



So, it is important for forensic investigators to understand how to examine Linux systems, and use Linux workstation for forensic examination



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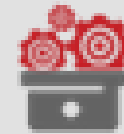


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# Collecting Volatile Data

- ❑ Volatile data is lost when a machine is **turned off/powered-down**
- ❑ However, during forensic investigation, investigators need to collect this data to construct a **timeline analysis** of the incident that occurred



## Volatile information includes:



Hostname, date & time, and time zone



Uptime



Network information



Open ports



Open files



Mounted file system information



Loaded kernel modules



User events



Running processes



Swap areas and Disk partition information



Kernel messages



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# Collecting Hostname, Date, and Time

- ❑ Identify the computer name using the **hostname** command  
Command:  
**hostname**
- ❑ This command can be useful while examining logs and network traffic



```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# hostname  
ubuntu  
root@ubuntu:~#
```

- ❑ Check the **date and time** of the machine to build a proper timeline of events  
Command:  
**date**  
**cat /etc/timezone**



```
root@ubuntu: /home/investigator  
File Edit View Search Terminal Help  
root@ubuntu:/home/investigator# date  
Wed Apr 22 23:53:43 PDT 2020  
root@ubuntu:/home/investigator# cat /etc/timezone  
America/Los_Angeles  
root@ubuntu:/home/investigator#
```

# Collecting Uptime Data



- ❑ The **uptime** command in Linux system displays how long the system has been running since the last restart



- ❑ This command also returns the current time, number of presently logged-in users, system load averages, etc.

Command:

**uptime**

```
root@ubuntu: /home/investigator
File Edit View Search Terminal Help

top - 07:00:28 up 48 wls, 1 user, load averages: 0.12, 0.14, 0.14
tasks: 328 total, 2 running, 295 sleeping, 3 stopped, 0 zombie
Mem(s): 19.5 vs, 4.0 fr, 0.0 ni, 76.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
Mem Mem: 2005964 total, 164348 free, 1070400 used, 763000 buff/cache
Mem Swap: 969960 total, 484168 free, 283792 used, 755996 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S  %CPU  %MEM     time+  COMMAND
 1536 investl+  20   0 2934200 170092 41712 R   16.2   8.5  0:52.20 /usr/bin/g+
21241 investl+  20   0 624212  31300 25000 S    3.6   1.6  0:09.11 /usr/bin/g+
1392 investl+  20   0 471112  44412 13900 S    2.0   2.2  0:23.55 /usr/lib/x+
  11 root      20   0      0      0      0 I    0.3   0.0  0:01.21 [rcu_sched]
 242 root      20   0      0      0      0 I    0.3   0.0  0:02.09 [kworker/0+
 429 root      20   0      0      0      0 I    0.3   0.0  0:01.41 [irq/ti-ve+
1397 investl+  20   0  51120  3400  1856 S    0.3   0.2  0:00.56 /usr/bin/d+
20000 investl+  20   0 662752 38272 27900 S    0.3   1.9  0:07.00 /usr/lib/g+
   1 root      20   0 225704  4900  4104 S    0.0   0.1  0:07.22 /lib/system+
   2 root      20   0      0      0      0 S    0.0   0.0  0:00.00 [kthread0]
   3 root      0 -20      0      0      0 I    0.0   0.0  0:00.00 [rcu_gp]
   4 root      0 -20      0      0      0 I    0.0   0.0  0:00.00 [rcu_par_g+
   6 root      0 -20      0      0      0 I    0.0   0.0  0:00.00 [kworker/0+
   7 root      20   0      0      0      0 I    0.0   0.0  0:01.59 [kworker/0+
   9 root      0 -20      0      0      0 I    0.0   0.0  0:00.00 [m_percpu+
  10 root      20   0      0      0      0 S    0.0   0.0  0:00.00 [ksoftirqd+]
```



```
root@ubuntu: /home/investigator
File Edit View Search Terminal Help

top - 07:06:28 up 40 min, 1 user, load average: 0.12, 0.14, 0.14
Tasks: 326 total, 2 running, 255 sleeping, 1 stopped, 0 zombie
%Cpu(s): 19.5 us, 4.0 sy, 0.0 ni, 76.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 2005964 total, 164548 free, 1078400 used, 763016 buff/cache
KiB Swap: 969960 total, 686168 free, 283792 used. 755996 avail Mem

  PID USER      PR  NI   VIRT   RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
 1536 investi+ 20   0 2954208 170092 41712 R 16.2   8.5   0:52.28 /usr/bin/g+
21241 investi+ 20   0 624212  31380 25060 S  3.6   1.6   0:00.11 /usr/bin/g+
 1392 investi+ 20   0 471512  44412 13960 S  2.0   2.2   0:23.55 /usr/lib/x+
   11 root      20   0      0       0      0  I  0.3   0.0   0:01.21 [rcu_sched]
  242 root      20   0      0       0      0  I  0.3   0.0   0:02.89 [kworker/0+
  429 root     -51   0      0       0      0  S  0.3   0.0   0:01.41 [irq/16-vm+
 1397 investi+ 20   0   51120   3440  1856 S  0.3   0.2   0:00.56 /usr/bin/d+
20688 investi+ 20   0 802732  38272 27980 S  0.3   1.9   0:07.88 /usr/lib/g+
   1 root      20   0 225784   6984  4144 S  0.0   0.3   0:07.22 /lib/syste+
   2 root      20   0      0       0      0  S  0.0   0.0   0:00.00 [kthreadd]
   3 root       0 -20      0       0      0  I  0.0   0.0   0:00.00 [rcu_gp]
   4 root       0 -20      0       0      0  I  0.0   0.0   0:00.00 [rcu_par_g+
   6 root       0 -20      0       0      0  I  0.0   0.0   0:00.00 [kworker/0+
   7 root      20   0      0       0      0  I  0.0   0.0   0:01.59 [kworker/0+
   9 root       0 -20      0       0      0  I  0.0   0.0   0:00.00 [mm_percpu+
  10 root      20   0      0       0      0  S  0.0   0.0   0:00.69 [ksoftirqd+
```

Result obtained upon running uptime command without any options

# Collecting Network Information

- ❑ The following syntax displays all **Network Interface Controllers** (NICs) and associated IP addresses associated with them

Syntax:

**ip addr show**

Note:

- lo, ens33 are NICs
- State **UNKNOWN – NIC** is operational but there is no connection
- State **UP – NIC** is operational and there is a connection

```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# ip addr show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:be:0f:09 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.135.155/24 brd 192.168.135.255 scope global dynamic noprefixroute ens33  
        valid_lft 1172sec preferred_lft 1172sec  
    inet6 fe80::401d:210e:9850:a56/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
root@ubuntu:~#
```





```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# ifconfig lo  
lo: flags=329<UP,LOOPBACK,RUNNING,PROMISC> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 137887 bytes 6222656 (6.2 MB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 137887 bytes 6222656 (6.2 MB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
root@ubuntu:~#
```

**Disabling promiscuous mode using ifconfig command**

```
root@ubuntu: /home/Investigator
File Edit View Search Terminal Help
root@ubuntu: /home/Investigator# netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State           I-Node    Path
unix  2      [ ]     DGRAM          State           I-Node    Path
unix  2      [ ]     DGRAM          State           I-Node    Path
unix  2      [ ]     DGRAM          State           I-Node    Path
unix  25     [ ]     DGRAM          State           I-Node    Path
unix  2      [ ]     DGRAM          State           I-Node    Path
unix  3      [ ]     DGRAM          State           I-Node    Path
unix  10     [ ]     DGRAM          State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     DGRAM          State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
unix  3      [ ]     STREAM  CONNECTED      State           I-Node    Path
```

Execution of netstat command

```
root@ubuntu: /home/Investigator
File Edit View Search Terminal Help
root@ubuntu: /home/Investigator# netstat -i
Kernel Interface table
Iface    MTU    RX-OK RX-ERR RX-DRP RX-OVR    TX-OK TX-ERR TX-DRP TX-OVR Flg
ens33    1500   55785 0      0 0      30553 0      0      0 BMRU
lo       65536  18197 0      0 0      18197 0      0      0 LRU
root@ubuntu: /home/Investigator#
```

Execution of netstat command with -i option

# Viewing Network Routing Tables

- ❑ The **netstat** command can also be used to print **routing tables**

Command:

**netstat -rn**

- r displays the kernel IP routing table
- n displays the numerical addresses

```
root@ubuntu: /home/investigator
File Edit View Search Terminal Help
root@ubuntu: /home/investigator# netstat -rn
Kernel IP routing table
Destination Gateway Genmask Flags MSS Window irtt Iface
0.0.0.0 192.168.135.1 0.0.0.0 UG 0 0 0 ens33
169.254.0.0 0.0.0.0 255.255.0.0 U 0 0 0 ens33
192.168.135.0 0.0.0.0 255.255.255.0 U 0 0 0 ens33
root@ubuntu: /home/investigator#
```

```
root@ubuntu: /home/investigator
File Edit View Search Terminal Help
root@ubuntu: /home/investigator# ip r
default via 192.168.135.1 dev ens33 proto dhcp metric 100
169.254.0.0/16 dev ens33 scope link metric 1000
192.168.135.0/24 dev ens33 proto kernel scope link src 192.168.135.130 metric 300
root@ubuntu: /home/investigator#
```

- ❑ In Linux, the **routing table** provides information on the process of forwarding TCP/IP data packets

Command:

**ip r**



# Collecting Open Port Information



Open ports can be vulnerable, and attackers use them to exploit a machine or a server



To gather information on open ports from the system, use the following commands:

❖ For **TCP** port connections:

Syntax:

```
nmap -sT localhost
```

❖ For **UDP** port connections:

Syntax:

```
nmap -sU localhost
```

## Displaying TCP port connections

```
root@ubuntu:/home/investigator
File Edit View Search Terminal Help
root@ubuntu:/home/investigator# nmap -sT localhost

Starting Nmap 7.60 ( https://nmap.org ) at 2020-04-23 04:38 PM
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000000s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
root@ubuntu:/home/investigator#
```

## Displaying UDP port connections

```
root@ubuntu:/home/investigator
File Edit View Search Terminal Help
root@ubuntu:/home/investigator# nmap -sU localhost

Starting Nmap 7.60 ( https://nmap.org ) at 2020-04-23 04:47 PM
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000000s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
68/udp    open|filtered dhcp
631/udp   open|filtered lpp
1351/udp   open|filtered avastconf

Nmap done: 1 IP address (1 host up) scanned in 2.72 seconds
root@ubuntu:/home/investigator#
```



```
root@ubuntu: /home/investigator
File Edit View Search Terminal Help
root@ubuntu: /home/investigator# nmap -sT localhost

Starting Nmap 7.60 ( https://nmap.org ) at 2020-04-23 04:38 PDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000074s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
631/tcp   open ipp

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

Execution of nmap command to check for TCP port connections

# Collecting Data on Open Files

- ❑ You can run **lsdf** command to list all open files as well as the active processes that opened them on the system

Command:

**lsdf**

- ❑ To list the open files for the user currently logged into the system

Command:

**lsdf -u <user\_name>**

```
root@ubuntu:~# lsdf / more
lsdf: WARNING: can't stat() fuse.gvfs-fuse File system /run/user/1000/gvfs:
Output information may be incomplete.

```

| COMMAND | PID | TID | USER | FD  | TYPE | DEVICE | SIZE/OFF | MODE   | NAME                                          |
|---------|-----|-----|------|-----|------|--------|----------|--------|-----------------------------------------------|
| systemd | 1   |     | root | cmd | DIR  | 8,1    | 4096     | 2      | /                                             |
| systemd | 1   |     | root | rtd | DIR  | 8,1    | 4096     | 2      | /                                             |
| systemd | 1   |     | root | txt | RGC  | 8,1    | 1022132  | 071799 | /lib/systemd/systemd                          |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 1700792  | 001843 | /lib/x86_64-linux-gnu/libc-2.27.so            |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 125816   | 057004 | /lib/x86_64-linux-gnu/libudev.so.1.0.0        |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 04032    | 001821 | /lib/x86_64-linux-gnu/libapp-error.so.0.22.0  |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 43204    | 001832 | /lib/x86_64-linux-gnu/libjem-0.0.3.0.1        |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 34072    | 272191 | /usr/lib/x86_64-linux-gnu/libc-glibc.so.0     |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 432640   | 001802 | /lib/x86_64-linux-gnu/libcrypt.so.1.0.0       |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 10680    | 001768 | /lib/x86_64-linux-gnu/libcrti.so.1.1.0        |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 18712    | 001783 | /lib/x86_64-linux-gnu/libcrtbegin.so.0.0.0    |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 27112    | 000830 | /lib/x86_64-linux-gnu/libcrtend.so.1.1.0      |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 14560    | 001803 | /lib/x86_64-linux-gnu/libc-2.27.so            |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 404024   | 001902 | /lib/x86_64-linux-gnu/libcpre.so.1.1.0        |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 144976   | 001913 | /lib/x86_64-linux-gnu/libcpre.so.1.1.0        |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 112672   | 272033 | /usr/lib/x86_64-linux-gnu/libc-2.27.so        |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 153004   | 001840 | /lib/x86_64-linux-gnu/libc-2.27.so            |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 200072   | 001827 | /lib/x86_64-linux-gnu/libc-2.27.so            |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 27000    | 272199 | /usr/lib/x86_64-linux-gnu/libc-glibc.so.0.0.0 |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 1150004  | 001828 | /lib/x86_64-linux-gnu/libcrypt.so.2.0.2.0     |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 22704    | 001785 | /lib/x86_64-linux-gnu/libc-2.27.so            |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 310040   | 001793 | /lib/x86_64-linux-gnu/libcrypt.so.1.1.0       |
| systemd | 1   |     | root | mem | RGC  | 8,1    | 11232    | 001758 | /lib/x86_64-linux-gnu/libc-2.27.so            |

# Viewing Running Processes in the System



- ❑ Run the **ps** command to view the processes running on the system
- ❑ It provides a **snapshot** of the **current processes** along with detailed information, such as the **user id**, **CPU usage**, **memory usage**, and **command name**
- ❑ Check the **process tree** to determine any suspicious **child processes** and **dependencies**



```
root@ubuntu:~# ps auxww
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 160240 6028 ?        Ss   11:52   0:04 /sbin/init auto noprompt
root         2  0.0  0.0      0     0 ?        S    11:52   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        I<   11:52   0:00 [rcu_gp]
root         4  0.0  0.0      0     0 ?        I<   11:52   0:00 [rcu_par_gp]
root         6  0.0  0.0      0     0 ?        I<   11:52   0:00 [kworker/0:0m-kb]
root         9  0.0  0.0      0     0 ?        I<   11:52   0:00 [nn_percpu_wq]
root        10  0.0  0.0      0     0 ?        S    11:52   0:01 [ksoftirqd/0]
root        11  0.0  0.0      0     0 ?        I    11:52   0:01 [rcu_sched]
root        12  0.0  0.0      0     0 ?        S    11:52   0:00 [nigratn/0]
root        13  0.0  0.0      0     0 ?        S    11:52   0:00 [idle_inject/0]
root        14  0.0  0.0      0     0 ?        S    11:52   0:00 [cpuhp/0]
root        15  0.0  0.0      0     0 ?        S    11:52   0:00 [kdevtmpfs]
root        16  0.0  0.0      0     0 ?        I<   11:52   0:00 [netns]
root        17  0.0  0.0      0     0 ?        S    11:52   0:00 [rcu_tasks_kthre]
root        18  0.0  0.0      0     0 ?        S    11:52   0:00 [kauditd]
root        19  0.0  0.0      0     0 ?        S    11:52   0:00 [khungtaskd]
root        20  0.0  0.0      0     0 ?        S    11:52   0:00 [oom_reaper]
root        21  0.0  0.0      0     0 ?        I<   11:52   0:00 [writeback]
root        22  0.0  0.0      0     0 ?        S    11:52   0:00 [kcompactd0]
root        23  0.0  0.0      0     0 ?        SN   11:52   0:00 [ksmd]
root        24  0.0  0.0      0     0 ?        SN   11:52   0:00 [khugepaged]
```

# Collecting Non-volatile Data

- ❑ The state of non-volatile data **does not change** when the machine is turned off
- ❑ During forensic investigation, the investigators should **collect non-volatile information** such as system information, user login history, system logs, and hidden files to construct timeline analysis of the incident that occurred

## Non-volatile data includes the following:

|   |                                             |    |                                                          |
|---|---------------------------------------------|----|----------------------------------------------------------|
| 1 | System Information                          | 7  | User history file                                        |
| 2 | Kernel information                          | 8  | Hidden files and directories                             |
| 3 | User accounts                               | 9  | Suspicious information                                   |
| 4 | Currently logged-in users and login history | 10 | File signatures                                          |
| 5 | System logs                                 | 11 | File information obtained using file and strings command |
| 6 | Linux log files                             | 12 | Writable files obtained using the find command           |



# Collecting Kernel Information

- ❑ Use the following **commands** to check the Linux **kernel version** on a system:

**uname -r**

(or)

**cat /proc/version**

(or)

**hostnamectl | grep Kernel**



```
root@ubuntu: /home/Investigator
File Edit View Search Terminal Help
root@ubuntu:/home/Investigator# uname -r
5.3.0-28-generic
root@ubuntu:/home/Investigator# cat /proc/version
Linux version 5.3.0-28-generic (bulldog@lcy01-amd64-009) (gcc version 7.4.0
(Ubuntu 7.4.0-1ubuntu1~18.04.1)) #30-18.04.1-Ubuntu SMP Fri Jan 17 06:14:
09 UTC 2020
root@ubuntu:/home/Investigator# hostnamectl | grep Kernel
kernel: Linux 5.3.0-28-generic
root@ubuntu:/home/Investigator#
```



# Collecting User Account Information



The `/etc/passwd` file running on a Linux system stores **local user** account information



Analyzing the `/etc/passwd` file allows the investigator to view **the user accounts** on the system



**Command:**

`cat /etc/passwd`



Command given to list only **usernames** in the output

`cut -d: -f1 /etc/passwd`



Each line in the output represents the **login information** of a single **user** and includes seven fields separated by colon (:)



You can observe the following about the output format of the `/etc/passwd` file by analyzing the first entry in the screenshot:

`root` – Username

`x` – Password ('x' denotes encrypted)

`0` – User ID ('0' is reserved for root)

`0` – Group ID

`root` – User ID information

`/root` – Home directory

`/bin/bash` – Absolute path to the user's login shell

```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# cat /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mailng List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin  
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd/netif:/usr/sbin/nologin  
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd/resolve:/usr/sbin/nologin  
syslog:x:102:106::/home/syslog:/usr/sbin/nologin  
messagebus:x:103:107::/nonexistent:/usr/sbin/nologin
```





# Collecting Currently Logged-in Users and Login History Information

- ❑ To find the currently **logged in user** in the system

Command:

**W**

```
root@ubuntu: /home/investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu: /home/investigator/Desktop# w
18:52:24 up 4:14, 1 user, load averages: 0.00, 0.01, 0.00
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU  WHAT
Investig  :0        :0            04:18    Tadm?   1:27   0.01s /usr/lib/gdm
root@ubuntu: /home/investigator/Desktop#
```

- ❑ The log file **/var/log/wtmp** maintains information about the user login history, system reboot time and system status
- ❑ The **last** command pulls the login history from the **wtmp** log file

Command:

**last -f /var/log/wtmp**



```
root@ubuntu: /home/investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu: /home/investigator/Desktop# last -f /var/log/wtmp
Investig  :0        :0            Tue Apr 28 04:18    gone - no logout
reboot    system boot  5.3.0-28-generic Tue Apr 28 04:37    still running
Investig  :0        :0            Tue Apr 28 04:02 - 04:36 (00:33)
reboot    system boot  5.3.0-28-generic Tue Apr 28 03:58 - 04:37 (00:39)
Investig  :0        :0            Mon Apr 27 22:55 - down (01:04)
reboot    system boot  5.3.0-28-generic Mon Apr 27 22:54 - 00:00 (01:05)
Investig  :0        :0            Mon Apr 27 06:05 - down (03:40)
reboot    system boot  5.3.0-28-generic Mon Apr 27 06:04 - 09:55 (03:46)
Investig  :0        :0            Mon Apr 27 04:57 - down (01:07)
reboot    system boot  5.3.0-28-generic Sun Apr 26 23:32 - 00:04 (00:31)
Investig  :0        :0            Sun Apr 26 22:39 - 23:11 (00:32)
reboot    system boot  5.3.0-28-generic Sun Apr 26 22:37 - 23:11 (00:34)
Investig  :0        :0            Sun Apr 26 18:19 - down (00:29)
reboot    system boot  5.3.0-28-generic Sun Apr 26 18:17 - 18:49 (00:31)
Investig  :0        :0            Thu Apr 23 23:13 - down (00:55)
reboot    system boot  5.3.0-28-generic Thu Apr 23 22:56 - 00:00 (00:11)
Investig  :0        :0            Thu Apr 23 22:42 - down (00:13)
reboot    system boot  5.3.0-28-generic Thu Apr 23 22:36 - 22:56 (00:20)
Investig  :0        :0            Thu Apr 23 12:26 - 12:52 (00:25)
Investig  :0        :0            Wed Apr 22 22:25 - 12:24 (13:58)
reboot    system boot  5.3.0-28-generic Wed Apr 22 22:24 - 12:52 (14:27)
Investig  :0        :0            Tue Apr 21 23:26 - 11:39 (12:12)
reboot    system boot  5.3.0-28-generic Tue Apr 21 23:26 - 11:39 (12:13)
```

```
root@ubuntu: /home/investigator/Desktop

File Edit View Search Terminal Help
root@ubuntu:/home/investigator/Desktop# w
10:52:24 up 4:14, 1 user, load average: 0.00, 0.01, 0.00
USER      TTY      FROM          LOGIN@      IDLE   JCPU   PCPU WHAT
investig  :0        :0            04:38      ?xdm?    1:27   0.01s /usr/lib/gdm
root@ubuntu:/home/investigator/Desktop#
```

Running w command

```
root@ubuntu: /home/Investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu: /home/Investigator/Desktop# last -f /var/log/wtmp
investig :0 :0 Tue Apr 28 04:38 gone - no logout
reboot system boot 5.3.0-28-generic Tue Apr 28 04:37 still running
investig :0 :0 Tue Apr 28 04:02 - 04:36 (00:33)
reboot system boot 5.3.0-28-generic Tue Apr 28 03:58 - 04:37 (00:39)
investig :0 :0 Mon Apr 27 22:55 - down (01:04)
reboot system boot 5.3.0-28-generic Mon Apr 27 22:54 - 00:00 (01:05)
investig :0 :0 Mon Apr 27 06:05 - down (03:46)
reboot system boot 5.3.0-28-generic Mon Apr 27 06:04 - 09:51 (03:46)
investig :0 :0 Mon Apr 27 04:57 - down (01:07)
reboot system boot 5.3.0-28-generic Sun Apr 26 23:32 - 06:04 (06:31)
investig :0 :0 Sun Apr 26 22:39 - 23:11 (00:32)
reboot system boot 5.3.0-28-generic Sun Apr 26 22:37 - 23:11 (00:34)
investig :0 :0 Sun Apr 26 10:19 - down (00:29)
reboot system boot 5.3.0-28-generic Sun Apr 26 10:17 - 10:49 (00:31)
investig :0 :0 Thu Apr 23 23:13 - down (08:55)
reboot system boot 5.3.0-28-generic Thu Apr 23 22:56 - 08:08 (09:11)
investig :0 :0 Thu Apr 23 22:42 - down (00:13)
reboot system boot 5.3.0-28-generic Thu Apr 23 22:36 - 22:56 (00:20)
investig :0 :0 Thu Apr 23 12:26 - 12:52 (00:25)
investig :0 :0 Wed Apr 22 22:25 - 12:24 (13:58)
reboot system boot 5.3.0-28-generic Wed Apr 22 22:24 - 12:52 (14:27)
investig :0 :0 Tue Apr 21 23:26 - 11:39 (12:12)
reboot system boot 5.3.0-28-generic Tue Apr 21 23:26 - 11:39 (12:13)
```



# Collecting System Logs Data

- ❑ On a Linux machine, the system logs are located in the directory **/var/log/syslog**
- ❑ The **syslog configuration file** stores system messages from logging facility and collects data logs of various programs and services, including the kernel

Command:

```
cat /var/log/syslog
```



```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# cat /var/log/syslog  
May 5 04:22:11 ubuntu rsyslogd: [origin software="rsyslogd" swversion="8.32.0"  
x-pid="683" x-info="http://www.rsyslog.com"] rsyslogd was HUPed  
May 5 04:22:14 ubuntu anacron[673]: Job 'cron.daily' terminated  
May 5 04:22:14 ubuntu anacron[673]: Normal exit (1 job run)  
May 5 04:23:26 ubuntu systemd[1]: Created slice User Slice of Investigator.  
May 5 04:23:26 ubuntu systemd[1]: Starting User Manager for UID 1000...  
May 5 04:23:26 ubuntu systemd[1]: Started Session 2 of user Investigator.  
May 5 04:23:26 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent and  
passphrase cache (restricted).  
May 5 04:23:26 ubuntu systemd[1536]: Starting D-Bus User Message Bus Socket.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on REST API socket for snapd use  
r session agent.  
May 5 04:23:27 ubuntu systemd[1536]: Started Pending report trigger for Ubuntu  
Report.  
May 5 04:23:27 ubuntu systemd[1536]: Reached target Paths.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent and  
passphrase cache (access for web browsers).  
May 5 04:23:27 ubuntu systemd[1536]: Reached target Timers.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent and  
passphrase cache.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent (ss  
h-agent emulation).  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG network certificate man  
agement daemon.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on D-Bus User Message Bus Socket
```

- ❑ Analyzing **Linux kernel logs** located at **/var/log/kern.log** can be helpful for troubleshooting custom kernels

Command:

```
cat /var/log/kern.log
```

```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# cat /var/log/kern.log  
May 4 11:53:15 ubuntu kernel: [10835.075027] usb 2-2.1: USB disconnect, device  
number 4  
May 4 23:19:46 ubuntu kernel: [10835.615227] e1000: eno33 N3C Link is Down  
May 4 23:19:46 ubuntu kernel: [10835.616015] usb 2-2.1: new full-speed USB devi  
ce number 5 using uhci_hcd  
May 4 23:19:46 ubuntu kernel: [10835.665146] usb 1-1: reset high-speed USB devi  
ce number 2 using ehci-pci  
May 4 23:19:47 ubuntu kernel: [10835.766583] usb 2-2.1: config 1 Interface 1 al  
tsetting 0 endpoint 0x3 has wMaxPacketSize 0, skipping  
May 4 23:19:47 ubuntu kernel: [10835.766585] usb 2-2.1: config 1 Interface 1 al  
tsetting 0 endpoint 0x3 has wMaxPacketSize 0, skipping  
May 4 23:19:47 ubuntu kernel: [10835.778739] usb 2-2.1: New USB device found, 1  
bVendor=0e0f, 1bProduct=0008, 1bcdDevice= 1.00  
May 4 23:19:47 ubuntu kernel: [10835.778751] usb 2-2.1: New USB device strings:  
 0xff=1, Product=2, SerialNumber=3  
May 4 23:19:47 ubuntu kernel: [10835.778785] usb 2-2.1: Product: Virtual Bluetoo  
oth Adapter  
May 4 23:19:47 ubuntu kernel: [10835.778787] usb 2-2.1: Manufacturer: VMware  
May 4 23:19:47 ubuntu kernel: [10835.778787] usb 2-2.1: SerialNumber: 000658268  
328  
May 4 23:19:48 ubuntu kernel: [10837.634747] e1000: eno33 N3C Link is Up 1000 M  
bps Full Duplex, Flow Control: None  
May 4 23:19:48 ubuntu kernel: [10837.641505] IPv6: ADDRCONF(NETDEV_CHANGE): eno  
33: link becomes ready
```

```
root@ubuntu: ~  
File Edit View Search Terminal Help  
root@ubuntu:~# cat /var/log/syslog  
May 5 04:22:11 ubuntu rsyslogd: [origin software="rsyslogd" swVersion="8.32.0"  
x-pid="683" x-info="http://www.rsyslog.com"] rsyslogd was HUPed  
May 5 04:22:14 ubuntu anacron[673]: Job 'cron.daily' terminated  
May 5 04:22:14 ubuntu anacron[673]: Normal exit (1 job run)  
May 5 04:23:26 ubuntu systemd[1]: Created slice User Slice of investigator.  
May 5 04:23:26 ubuntu systemd[1]: Starting User Manager for UID 1000...  
May 5 04:23:26 ubuntu systemd[1]: Started Session 2 of user investigator.  
May 5 04:23:26 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent and  
passphrase cache (restricted).  
May 5 04:23:26 ubuntu systemd[1536]: Starting D-Bus User Message Bus Socket.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on REST API socket for snapd use  
r session agent.  
May 5 04:23:27 ubuntu systemd[1536]: Started Pending report trigger for Ubuntu  
Report.  
May 5 04:23:27 ubuntu systemd[1536]: Reached target Paths.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent and  
passphrase cache (access for web browsers).  
May 5 04:23:27 ubuntu systemd[1536]: Reached target Timers.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent and  
passphrase cache.  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG cryptographic agent (ss  
h-agent emulation).  
May 5 04:23:27 ubuntu systemd[1536]: Listening on GnuPG network certificate man  
agement daemon.
```

Examining syslog file using cat command



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## Linux Log Files

| Log Location                     | Content Description                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------|
| <code>/var/log/auth.log</code>   | System authorization information, including user logins and authentication mechanism    |
| <code>/var/log/kern.log</code>   | Initialization of kernels, kernel errors or informational messages sent from the kernel |
| <code>/var/log/faillog</code>    | Failed user login attempts                                                              |
| <code>/var/log/lpr.log</code>    | Printer logs                                                                            |
| <code>/var/log/mail.*</code>     | All mail server message logs                                                            |
| <code>/var/log/mysql.*</code>    | All MySQL server logs                                                                   |
| <code>/var/log/apache2/*</code>  | All Apache web server logs                                                              |
| <code>/var/log/apport.log</code> | Application crash report/log                                                            |
| <code>/var/log/lighttpd/*</code> | Lighttpd web server log files directory                                                 |
| <code>/var/log/daemon.log</code> | Running services, such as squid and ntpd                                                |
| <code>/var/log/debug</code>      | Debugging log messages                                                                  |
| <code>/var/log/dpkg.log</code>   | Package installation or removal logs                                                    |



# Module Flow

Understand Volatile  
and Non-Volatile  
Data in Linux

01



02

Analyze Filesystem  
Images Using The  
Sleuth Kit

Understand Mac  
Forensics

04

03

Demonstrate Memory  
Forensics



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# File System Analysis Using The Sleuth Kit: **fsstat**

- ❑ In Linux systems, the **fsstat** command provides information associated with the given file system
- ❑ The output of this command is filesystem-specific and consists of several information such as the file system type, volume ID, last mounted timestamps and last mounted directory

## Command:

```
fsstat -i <input_filetype>  
<filename.extension>
```

```
root@ubuntu: /home/investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu: /home/investigator/Desktop# fsstat -i raw Evidence.dd
FILE SYSTEM INFORMATION
.....
File System Type: Ext4
Volume Name:
Volume ID: c322aba24d168eab914b7ee5f9917521
Last Written at: 2020-05-01 08:14:49 (PDT)
Last Checked at: 2020-04-23 23:40:05 (PDT)
Last Mounted at: 2020-05-01 08:12:00 (PDT)
Unmounted properly
Last mounted on: /mnt/ext4
Source OS: Linux
Dynamic Structure
Compat Features: Journal, Ext Attributes, Resize Inode, Dir Index
InCompat Features: Filetype, Extents, 64bit, Flexible Block Groups,
Read Only Compat Features: Sparse Super, Large File, Huge File, Extra Inode Size
Journal ID: 00
Journal Tombstones: 0
```





```
root@ubuntu: /home/Investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu:/home/Investigator/Desktop# fsstat -i raw Evidence.dd
FILE SYSTEM INFORMATION
-----
File System Type: Ext4
Volume Name:
Volume ID: c322aba24d160eab914b7ee5f9917521

Last Written at: 2020-05-01 08:14:49 (PDT)
Last Checked at: 2020-04-23 23:40:05 (PDT)

Last Mounted at: 2020-05-01 08:12:00 (PDT)
Unmounted properly
Last mounted on: /mnt/ext4

Source OS: Linux
Dynamic Structure
Compat Features: Journal, Ext Attributes, Resize Inode, Dir Index
InCompat Features: Filetype, Extents, 64bit, Flexible Block Groups,
Read Only Compat Features: Sparse Super, Large File, Huge File, Extra Inode Size

Journal ID: 00
Journal Superblock: 0
```

Examining an image using fsstat command



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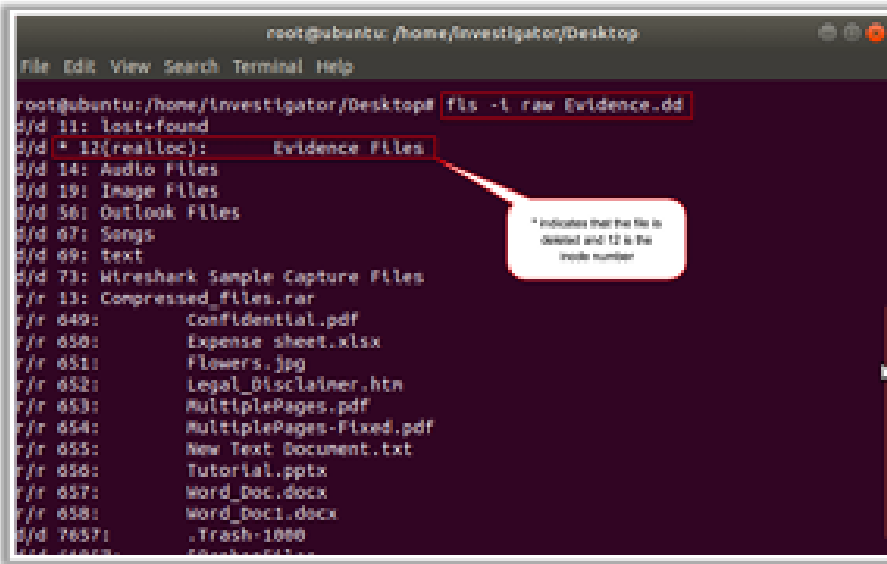
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# File System Analysis Using The Sleuth Kit: fls and istat

- ❑ Run the **fls** command to list the files and directories available in an image file
- ❑ This command is also useful to view **recently deleted files**

Command:

```
fls -i <image_type> <imagefile_name>
```

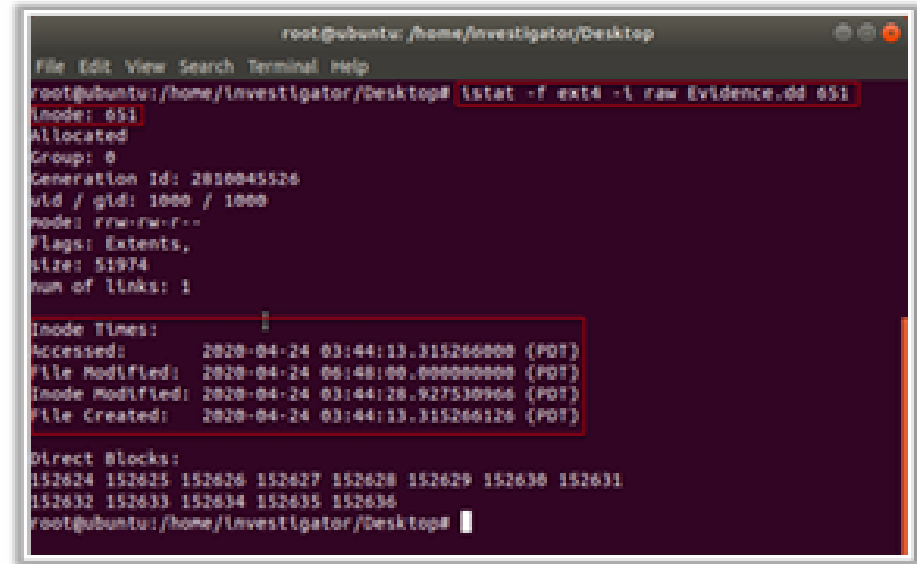


```
root@ubuntu: /home/Investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu: /home/Investigator/Desktop# fls -i raw Evidence.dd
d/d 11: lost+found
d/d * 12(realloc): Evidence Files
d/d 14: Audio Files
d/d 19: Image Files
d/d 56: Outlook Files
d/d 67: Songs
d/d 69: Text
d/d 73: Wireshark Sample Capture Files
r/r 13: Compressed_Files.rar
r/r 649: Confidential.pdf
r/r 650: Expense sheet.xlsx
r/r 651: Flowers.jpg
r/r 652: Legal_Disclaimer.htm
r/r 653: MultiplePages.pdf
r/r 654: MultiplePages-Fixed.pdf
r/r 655: New Text Document.txt
r/r 656: Tutorial.pptx
r/r 657: word_doc.docx
r/r 658: word_doc1.docx
d/d 7657: .Trash-1000
d/d 6557: 6557Files
```

- ❑ Use **istat** command that displays the metadata of a file, such as MAC times, file size, and file access permissions, by specifying a particular inode number

Command:

```
istat -f <fstype> -i <imgtype>
<imagefile_name> <inode_number>
```



```
root@ubuntu: /home/Investigator/Desktop
File Edit View Search Terminal Help
root@ubuntu: /home/Investigator/Desktop# istat -f ext4 -i raw Evidence.dd 651
Inode: 651
Allocated
Group: 0
Generation Id: 2830045526
uid / gid: 1000 / 1000
mode: rwxrwxr--
Flags: Extents,
size: 51974
num of links: 1

Inode Times:
Accessed: 2020-04-24 03:44:13.315260000 (PDT)
File Modified: 2020-04-24 04:48:00.000000000 (PDT)
Inode Modified: 2020-04-24 03:44:28.927530900 (PDT)
File Created: 2020-04-24 03:44:13.315260120 (PDT)

Direct Blocks:
152624 152625 152626 152627 152628 152629 152630 152631
152632 152633 152634 152635 152636
root@ubuntu: /home/Investigator/Desktop#
```

# Module Flow

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**Demonstrate Memory  
Forensics**

03



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# Memory Forensics: Introduction



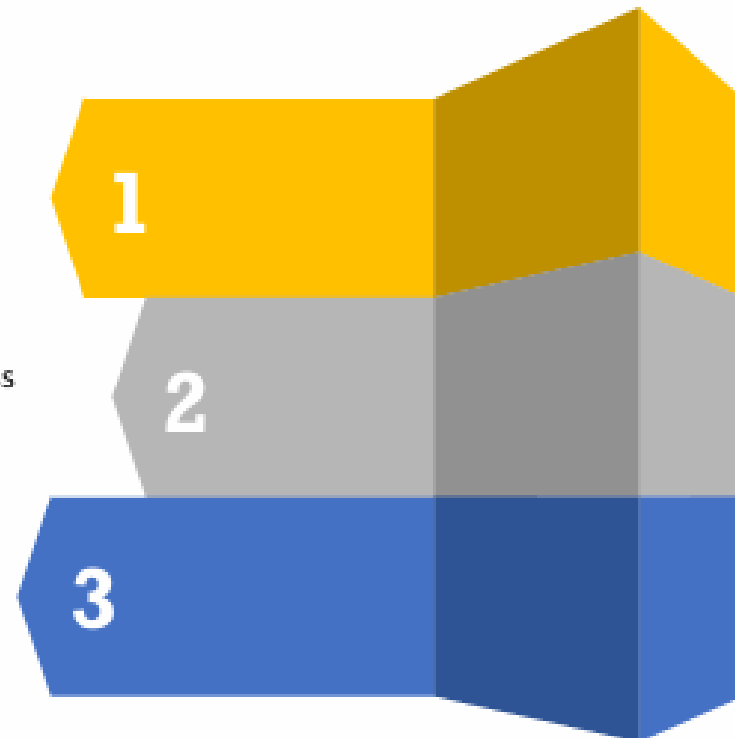
Memory forensics involves **forensic analysis of RAM dumps** captured from a running machine



Forensic analysis of **RAM dump** provides insights into processes running in the memory, network information, unauthorized access to the system, loaded modules, recently executed commands, injected code fragments, etc.



Such information can help the investigator **uncover malware attacks** or any other malicious behavior that has occurred on the target machine



**Note:** The investigator should proceed with the forensic examination based on the **information/events recorded** by the incident response team



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# Malware Analysis Using Volatility Framework



- ❑ After acquiring RAM dumps from the target machine, the investigator should analyze those dumps using tools such as **Volatility** to **identify** the occurrence of **malicious activity**
- ❑ To examine memory dumps using Volatility Framework, the investigator should **create a Linux profile** that matches the kernel version of the target RAM dump (which is used for analysis)



The **pslist** plugin lists all the **processes** that were running on the machine when the memory dump was captured

**Command:**

```
python vol.py --file=<file_name> --  
profile=<Linux_profile_name> linux_pslist
```

**Note:** In this case, the Linux profile is **Linux Ubuntu\_16.04 x64**

```
root@administrator-virtual-machine: /home/administrator/volatility
root@administrator-virtual-machine: /home/administrator/volatility# python vol.py --file=../ub
untu_ramdump.dd --profile=linuxubuntu_16_04x64 linux_pslist
Volatility Foundation Volatility Framework 2.6.1
Offset      Name      Start Time      PID      PPID      UID      GID
-----
0xfffffffff0000000 system    2020-05-04 08:11:23 UTC+0000      0      0      0
0xfffffffff0000000 kthreadd 2020-05-04 08:11:23 UTC+0000      0      0      0
0xfffffffff0000000 kworker/0:0 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 nm_percpu_wq 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 ksoftirqd/0 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 rcu_sched 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 rcu_bh 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 migration/0 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 watchdog/0 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 cpuhp/0 2020-05-04 08:11:23 UTC+0000      2      0      0
0xfffffffff0000000 cpuhp/1 2020-05-04 08:11:23 UTC+0000      2      0      0
```

# Carving Memory Dumps Using PhotoRec Tool



**PhotoRec** is an open-source tool that uses data carving techniques to **recover deleted files/lost data** from a drive or an image file



Memory dumps **contain volatile data** pertaining to logged on users, shared files, recently accessed media files and chats via social networks, accessed webpages, etc.



Identifying and extracting these files allows the forensic investigators to perform a more detailed investigation

## Carving data from the memory dump

- ❑ Run the **PhotoRec** tool and execute the below command  
Command:  
`photorec <Imagefile_name>`

```
root@administrator-virtual-machine: /home/administrator
root@administrator-virtual-machine: /home/administrator# photorec
ubuntu_randump.dd
```



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# Module Flow



# Introduction to Mac Forensics

1

The ever-increasing adoption of Mac systems has made them a primary target for malicious attacks

2

The advancement of **malware tools** and **lower availability** of security tools for MacOS-based systems has further expedited these threats

3

In order to identify an attack or prove guilt, **investigators require evidence** such as the presence of malware, unauthorized logging attempts, and connectivity to malicious servers and websites

4

Mac systems store all such **evidence data** in log files, directories, configurations, applications history, etc. and investigators need to extract these data and use them to create a timeline to figure out what happened

5

Therefore, to carry out an effective investigation, investigators should **possess** in-depth knowledge of the MacOS, its filesystem, libraries, and directories



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## Mac Forensics Data



### ❑ Detection of the System Version:

- View the **SystemVersion.plist** file located at  
/System/Library/CoreServices/SystemVersion.plist



❑ **Timestamp:**

- Use the command line input stat to find the timestamp of any file
- Usage: `stat [-Flnqrsx] [-F format] [-t timefmt] [file ...]`



- ❑ **Application bundles:**

- These are special directories that store application data, and are hidden from the user
- Analyze these bundles to **identify malware** or other **suspicious data**
- Evaluate the executable codes to check if something is wrong with the application



 **Finder:**

- It is the **default Mac application** that helps find specific files and folders
- It also helps in sorting in the required order

[illegible]

The terminal window shows the command `ls -l /Users/Hammerhead/Documents/Year1/DownloadUbuntu-16.04-desktop-amd64.iso` and its output:

```

-rw-r--r--  1 66666666  144K  1 Jun 25 16:40:10 2016
DownloadUbuntu-16.04-desktop-amd64.iso

```

```
Hammerhead — -bash — 106x33
Pazzero:~ Hammerhead$ cat /System/Library/CoreServices/SystemVersion.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>ProductBuildVersion</key>
  <string>15F34</string>
  <key>ProductCopyright</key>
  <string>1983-2016 Apple Inc.</string>
  <key>ProductName</key>
  <string>Mac OS X</string>
  <key>ProductUserVisibleVersion</key>
  <string>10.11.5</string>
  <key>ProductVersion</key>
  <string>10.11.5</string>
</dict>
</plist>
```

## Command:

```
stat [-FlLnqrsx] [-f format] [-t timefmt] [file ...]
```



A screenshot of a macOS terminal window titled "Hammerhead — -bash — 106x33". The prompt is "Pazzero:~ Hammerhead\$". The command entered is "stat -x /Users/Hammerhead/Documents/Vuze\ Downloads/kubuntu-16.04-desktop-amd64.iso". The output displays file metadata for the specified ISO file.

```
Pazzero:~ Hammerhead$ stat -x /Users/Hammerhead/Documents/Vuze\ Downloads/kubuntu-16.04-desktop-amd64.iso ]

  File: "/Users/Hammerhead/Documents/Vuze Downloads/kubuntu-16.04-desktop-amd64.iso"
  Size: 1520762880   FileType: Regular File
  Mode: (0644/-rw-r--r--)      Uid: ( 501/Hammerhead)  Gid: ( 20/  staff)
Device: 1,4   Inode: 6666764   Links: 1
Access: Sat Jun 25 19:02:32 2016
Modify: Fri Jun  3 14:54:49 2016
Change: Fri Jun  3 14:54:49 2016
```

**Checking Metadata associated with a file on a Mac system using stat command**



# Mac Log Files

| Log File                                          | Uses                                                 |
|---------------------------------------------------|------------------------------------------------------|
| <code>/var/log/crashreporter.log</code>           | Application crash history                            |
| <code>/var/log/cups/access_log</code>             | Printer connection information                       |
| <code>/var/log/cups/error_log</code>              | Printer connection information                       |
| <code>/var/log/daily.out</code>                   | Network interface history                            |
| <code>/var/log/samba/log.nmbd</code>              | Samba (Windows-based machine) connection information |
| <code>~/Library/Logs</code>                       | Application logs specific to Home directory          |
| <code>~/Library/Logs/iChatConnectionErrors</code> | iChat connection information                         |
| <code>~/Library/Logs/Sync</code>                  | Information of devices on .Mac syncing               |
| <code>/var/log/*</code>                           | Main folder for system log files                     |
| <code>/var/audit/*</code>                         | Audit logs                                           |
| <code>/var/log/install.log</code>                 | System and software update installation dates        |



# Mac Directories

| File name                                               | Location                                                                     |
|---------------------------------------------------------|------------------------------------------------------------------------------|
| Launch agent files                                      | /Library/LaunchAgents/* , /System/Library/LaunchAgents/*                     |
| Launch daemon files                                     | /Library/LaunchDaemons/* , /System/Library/LaunchDaemons/*                   |
| Startup item file                                       | /Library/StartupItems/* , /System/Library/StartupItems/*                     |
| Mac OS X jobs                                           | /usr/lib/cron/jobs/*                                                         |
| Cron tabs or scheduled jobs                             | /etc/crontab, /usr/lib/cron/tabs/*                                           |
| Wireless networks                                       | /Library/Preferences/SystemConfiguration/com.apple.airport.preferences.plist |
| User preference settings for applications and utilities | %%users.homedir%%/Library/Preferences/*                                      |
| Attached iDevices                                       | %%users.homedir%%/Library/Preferences/com.apple.iPod.plist                   |
| Social accounts                                         | %%users.homedir%%/Library/Accounts/Accounts3.sqlite                          |
| Trash directory                                         | %%users.homedir%%/.Trash/                                                    |
| Safari main folder                                      | %%users.homedir%%/Library/Safari/*                                           |
| Mozilla Firefox web browser                             | %%users.homedir%%/Library/Application Support/Firefox/*                      |
| Google Chrome web browser                               | %%users.homedir%%/Library/Application Support/Google/Chrome/*                |



## Parsing Metadata on Spotlight

01

**Spotlight** is a built-in indexing system on MacOS that creates indexes of all files/folders on the system and stores the metadata of every file/folder on the disk

03

The **store.db** database file is the Spotlight's central database repository. Each individual MacOS partition contains store.db file.



On MacOS, Spotlight can be accessed by pressing **Command + Space bar** keys

02

In digital forensic examination, the database information **parsed** from Spotlight retrieves details such as dates, last opened, and number of times an application or file is opened

04

# Mac Forensics Tools

**OS X Auditor**

<https://github.com>



**Recon Imager**

<https://suumurl.com>



**Memoryze for the Mac**

<https://www.fireeye.com>



**Stellar Data Recovery  
Professional for Mac**

<https://www.stellarinfo.com>



**F-Response**

<http://www.f-response.com>



**volafox**

<http://github.com>



**Volatility**

<http://www.volatilityfoundation.org>



**mac\_apr - macOS (and iOS)  
Artifact Parsing Tool**

<http://github.com>



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# Module Summary



- ➔ This module has discussed collecting volatile and non-volatile data in Linux
- ➔ It has discussed filesystem images analysis using The Sleuth Kit
- ➔ It has also discussed in detail the memory forensics using Volatility and PhotoRec
- ➔ Finally, this module ended with a detailed discussion on Mac forensics
- ➔ In the next module, we will discuss in detail on network forensics



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# REFERENCES

- **Digital Forensics Essentials – 2021 by EC-Council**